

Protocol



Integration of diverse bioactivity data into the Chemical Checker compound universe

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Abstract

Chemical signatures encode the physicochemical and structural properties of small molecules into numerical descriptors, forming the basis for chemical comparisons and search algorithms. The increasing availability of bioactivity data has improved compound representations to include biological effects (for example, induced gene expression changes), although bioactivity descriptors are often limited to a few well-documented molecules. To address this issue, we implemented a collection of deep neural networks able to leverage the experimentally determined bioactivity data associated to small molecules and infer the missing bioactivity signatures for any compound of interest. However, unlike static chemical descriptors, these bioactivity signatures dynamically evolve with new data and processing strategies. Here we present a computational protocol to modify or generate novel bioactivity spaces and signatures, describing the main steps needed to leverage diverse bioactivity data with the current knowledge, as catalogued in the Chemical Checker (CC; <https://chemicalchecker.org/>), using the predefined data curation pipeline. We illustrate the functioning of the protocol through four specific examples, including the incorporation of new compounds to an already existing bioactivity space, a change in the data preprocessing without altering the underlying experimental data and the creation of two novel bioactivity spaces from scratch, which are completed in under 9 h using graphics processing unit computing. Overall, this protocol offers a guideline for installing, testing and running the CC data integration approach on user-provided data, extending the annotation presented for a limited number of small molecules to a larger chemical landscape and generating novel bioactivity signatures.

Key points

- The Chemical Checker is a collection of bioactivity signatures for over 1 million small molecules. Data are organized into five levels of biological complexity and curation: from chemical properties to clinical outcomes and from raw data representing explicit knowledge to embedded signatures inferred from observed bioactivity patterns.
- This protocol describes how to integrate new data into the Chemical Checker universe of small molecules and generate novel customized bioactivity signatures.

Key references

- Duran-Frigola et al. *Nat. Biotechnol.* **38**, 1087–1096 (2020): <https://www.nature.com/articles/s41587-020-0502-7>
- Berton et al. *Nat. Commun.* **12**, 3932 (2021): <https://www.nature.com/articles/s41467-021-24150-4>

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Introduction

Small molecules are excellent tools for probing biological functions and continue to be the cornerstone of pharmaceutical research. Synthetic chemical compounds include a large variety of structural arrangements that, for the most part, have not been explored by natural evolution. Many pharmacological compounds are imperfect human inventions with suboptimal bioactivities that lack a clear connection to their chemical structure. Often, the only way to approach the biological characterization of a compound is to assume it will have comparable bioactivities to compounds with similar chemical properties. The so-called **similarity principle** has been the driving force of drug discovery efforts¹, and the measurement of compound similarities underlies most of the approaches used to explore the vast drug-like chemical space (estimated in the range of 10^{33} molecules²). To assess such similarities, molecules are usually characterized using **numerical fingerprints** or descriptors (for example, ref. 3), which encapsulate their main topological and physicochemical properties representing them in a format amenable for computational applications (for example, **quantitative structure–activity relationships**)⁴.

In the past years, the extensive gathering and release of bioactivity data have shown that the similarity principle applies beyond standard chemical features, reaching to functional properties. For instance, small molecules that show similar sensitivity profiles across human tumor cell lines or drugs that cause comparable side effects in patients tend to share mechanisms of action (MoA), even when they are structurally dissimilar^{5–7}. Thus, **bioactivity similarities** offer an alternative means to functionally characterize small molecules, potentially providing insights closer to clinical observations and surpassing what is expected from merely inspecting chemical analogs⁸.

The CC

We recently developed the **Chemical Checker** (CC; Fig. 1a), the largest collection so far of processed, harmonized and integrated bioactivity signatures; it contains over 1 million compounds, whose biological effects had been experimentally determined and curated by different data repositories⁹. Based on our understanding of drug action, we organized the data into 25 bioactivity spaces grouped in 5 levels of increasing biological complexity. The first level is the physicochemical and structural properties of small molecules (A) and the fifth level is the clinical effects they exert in human patients (E). In between, we collect the set of **biological targets** they bind (B), the perturbations they trigger in biological pathways (C) and the **phenotypic effects** they induce in cell-based assays (D).

Unfortunately, the availability of bioactivity data becomes scarcer and more difficult to obtain as biological complexity increases (that is, we have direct protein-binding data for ~630k small molecules but only ~6k compounds are annotated with clinical indications); a large proportion of compounds remain poorly characterized in terms of biological activity. Indeed, out of the 10^{33} synthetically accessible drug-like molecules², only a tiny fraction has experimentally measured chemical properties or bioactivities reported in public databases (that is, ~120M in PubChem and ~2.4M in ChEMBL as of October 2024).

To address this problem, we developed a collection of **deep neural networks to infer CC** bioactivity signatures for any compound of interest^{10,11}, based on the observation that the different bioactivity spaces are not completely orthogonal, and thus similarities of a given bioactivity type (for example, targets) can be transferred to other data kinds (for example, therapeutic indications). Moreover, we showed that the inferred bioactivity signatures are useful to navigate the chemical space in a biologically relevant manner and improve the performance of biophysics and physiology activity prediction compared with prediction tools based on chemistry-only classifiers¹⁰.

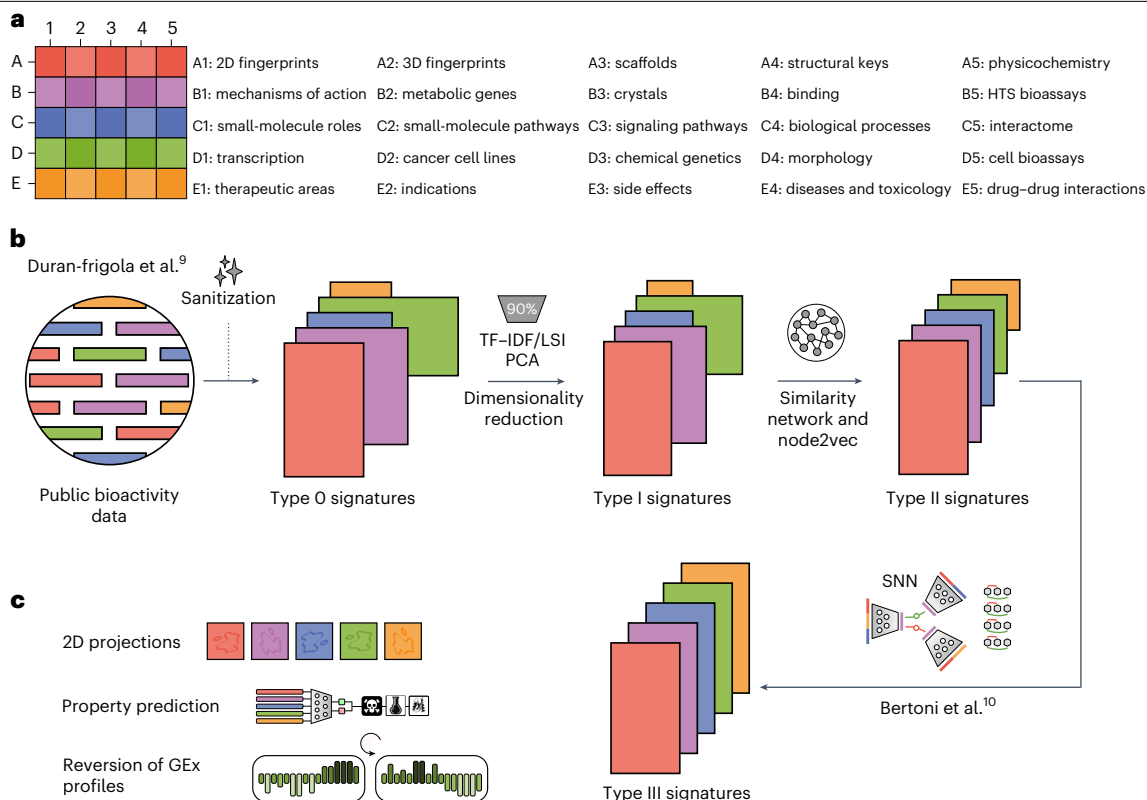


Fig. 1 | CC overview. **a**, Organization of the 25 CC spaces. HTS, high-throughput screening. **b**, The computational pipeline to build the CC and procedure to integrate bioactivity data in the CC universe. In brief, bioactivity data (here represented as numerical matrices—colored rectangles; rows are compounds and columns are biological features) is sanitized (type 0 signatures), compressed (type I signatures), harmonized (type II signatures) and integrated with all the

other data within the CC (type III signatures). **c**, Compound signatures are used in a variety of tasks, such as the characterization and visualization of large small-molecule libraries, the prediction of multiple properties (for example, toxicity, physicochemical properties, protein binding and so on) and the reversion of GEx profiles associated with specific disease states, among many others.

In general, we believe that if we can incorporate complex biological information associated to small molecules, in the form of bioactivity signatures, at the early stages of the drug discovery process, this would potentially overcome some of the problems associated with target-based approaches. For instance, using CC signatures, we were able to identify three compounds able to revert transcriptional signatures related to Alzheimer's disease *in vitro* and *in vivo*¹². Moreover, using CRISPR–Cas9 and short hairpin RNA perturbation experiments as templates, we also identified several small molecules that mimicked the phenotypic effects of biodrugs (for example, daclizumab, ustekizumab, cetuximab and so on), often through a different MoA that did not directly modulate the activity of their primary targets (for example, IL-2R, IL-12 and EGFR)⁹, and targeted cancer proteins thought to be undruggable¹⁰.

The original 25 CC spaces (Fig. 1a) comprise bioactivity data retrieved from public repositories that were processed through a unified data curation and integration pipeline (see Supplementary Table 1 for a detailed list of the bioactivity databases integrated in the CC). Compound bioactivities are expressed in a vector-like format (that is, signatures) and organized in increasing levels of abstraction: from raw data accounting for explicit knowledge (type 0 signatures) to inferred representations leveraging known bioactivity patterns (type III signatures). However, in the past years, new large-scale assays to capture biological responses to small-molecule perturbations, and more efficient computational strategies to process them, are constantly appearing (for example, refs. 13–15). Besides, there is an ever-growing number of laboratories and biotechnological companies that have their own in-house data that

they want to leverage, together with the bulk of known bioactivity information. Thus, in this study, we present the complete computational protocol to generate novel bioactivity spaces and signatures, describing the main steps needed to leverage diverse bioactivity data with the CC (type 0, type I, type II and type III signatures) using the predefined data curation and integration pipeline.

Overview of the CC data integration pipeline

The main input for the CC integration pipeline is a bioactivity data matrix encompassing compounds (rows, for example, InChIKeys) and biological features (columns, unique strings: for example, protein targets (B4), mechanisms of action (B1), clinical side effects (E3) and so on). Data might be categorical (for example, binary), discrete or continuous.

Type 0 signatures

The very first step in the herein presented CC processing framework is to produce a sufficiently curated and sanitized version of the raw bioactivity data (Fig. 1b). To achieve this, features (columns) that are extremely sparse (that is, occurring in few rows) or too prevalent (that is, occurring in almost all rows) are removed from the matrix and those compounds (rows) showing no feature occurrences are discarded. In addition, not-available and infinite values are processed (median-imputed and max/min values, respectively) and low-Shannon entropy features are trimmed if the maximum number of allowed features is surpassed. Overall, type 0 signatures represent explicit knowledge, allowing direct interpretation while having variable dimensionality.

Type I signatures

The second processing step implies a mild compression of type 0 signatures typically retaining 90% of the original variance and thereby reducing the number of vector dimensions (Fig. 1b). For discrete data, a term frequency–inverse document frequency (TF–IDF) transformation is used to weight features (columns) according to their information retrieval (that is, more frequent features are less informative and thus become underweighted). After that, a latent semantic indexing (LSI) dimensionality reduction technique is applied to the TF–IDF transformed data and LSI components explaining 90% of the original variance are kept to build type I signatures. For continuous data, the matrix is first scaled column-wise (median 0, median absolute deviation 1 and capped at ± 10) and then processed through a principal component analysis (PCA). Analogously to discrete data, PCA components explaining 90% of the original variance are kept to build type I signatures. Overall, type I signatures are better optimized for compound similarity calculations, although they no longer represent explicit knowledge and still have variable dimensionality.

Type II signatures

The main purpose of type II signatures is to retain similarities at type I signature level while harmonizing the length of vectors among CC spaces (that is, converting signatures of variable dimensionalities to 128-length vectors; Fig. 1b). First, a similarity network is built considering nearest neighbors (NN) at the type I signature level for each compound (small molecules represent the nodes in the graph and their similarities are the edges). Then, the node2vec algorithm is run to create an embedding for each molecule (node), encoding intrinsic relationships within the similarity graph. In this way and, although not directly interpretable, type II signatures capture explicit and implicit similarities in the data. Additionally, the fixed length (128) makes the concatenation of descriptors across bioactivity spaces cost-effective, while ensuring that each bioactivity space contributes equally to the overall representation.

Type III signatures

The last step in the processing framework is the generation of type III signatures for the input compounds and for all the molecules in the CC, which capture and infer the similarity of

the data at type I signature level (Fig. 1b). The approach hinges on the insight that different CC spaces are not completely orthogonal and may show correlation to some extent. In short, for each CC space, a Siamese neural network (SNN) is trained over 10 million triplets of compounds (anchor, positive and negative molecules sampled at type I signature level) representing these input molecules by concatenating all available type II signatures. After being validated, the SNN generates novel signatures (type III) that retain the observed similarities at type I signature level and accounts for compound data present in every other bioactivity space of the CC, allowing the inference of a bioactivity signature even in the absence of data for the molecule of interest. It is worth mentioning that the generation of type III signatures is usually the most computationally expensive step in the CC data integration pipeline, and strongly benefits from parallelization and graphics processing unit (GPU) computing (see the ‘Materials: hardware’ section).

Since most properties and similarities at the type III signature level are inferred, each of them comes together with a measure of confidence called ‘applicability’ (from 0 to 1). Applicability scores evaluate the distance to training-set signatures, the robustness of the predicted signatures and the expectancy a priori given the known data. Such scores are space dependent and enable compound ranking in terms of signature confidence: the higher the applicability score, the more confident the inference is. Reliable signature inferences in the 25 original bioactivity spaces showed applicability scores of ≥ 0.4 (ref. 10).

Additional remarks

The outlined pipeline includes numerous distinct steps, each possessing its own set of default options and parameters. While the user can modify the vast majority of parameters, the whole point of the computational pipeline—and where we put most emphasis—is in the full automation. In fact, the approach works end-to-end, and it is very robust to very different data shapes and types (sparse, long, wide, dense and so on). Thus, we recommend to use the by default parameters, but give the possibility to the user to adapt them if new uses of the CC signatures might arise. For additional details on such parameters and further clarification of the mentioned steps, we refer to the original publications^{9,10} and our regularly maintained GitLab repository (<https://gitlab.bnb.irbbarcelona.org/packages>).

Interpreting the results: diagnosis plots

To assess and monitor data flow at each step of the CC integration pipeline, we have designed several analyses to evaluate the results in a systematic manner. In brief, for each data matrix (compound signatures), a set of diagnosis plots is generated to illustrate the main characteristics of the data. This section includes a comprehensive explanation for each plot.

Examples for diagnosis plots are found in Extended Data Figs. 1, 3, 6 and 9. From left to right and from top to bottom:

- **Image:** a heat map representing the data matrix. Rows are compounds (keys, max number set to 100), columns are features and colors represent actual values from minimum (blue) to maximum (red)
- **Values distribution:** data values (x axis) and their distribution (y axis) from minimum (blue) to maximum (red)
- **Two-dimensional (2D) projection:** *t*-distributed stochastic neighbor embedding (*t*-SNE) representation of the data, colored by density. The maximum number of subsampled compounds included in the projection is 10,000
- **Euclidean distribution:** density distribution (y axis) of pairwise Euclidean distances (x axis). The maximum number of selected compound pairs is set to 10,000
- **Values by feature:** numerical values (y axis) of each feature (x axis)
- **Values by key:** numerical values (y axis) of each compound (key, x axis)
- **Redundancy:** for each signature in a nonredundant set of signatures (x axis), the number of compounds (keys, y axis) having the same exact signature
- **Cosine distribution:** density distribution (y axis) of pairwise cosine distances (x axis). The maximum number of selected compound pairs is set to 10,000

The full version of the diagnosis plots provided by CC comprises additional analysis and comparisons to other CC spaces. Examples of the full version are found in Fig. 4 and Extended Data Figs. 4, 7 and 10. The new plots in the full version are described below from left to right, top to bottom:

- **Confidence**: density plot of normalized confidence values for the signatures. Mean values above 0.6 indicate reliable signatures
- **2D projection colored by confidence value**: *t*-SNE representation of the data, colored by their assigned confidence values
- **CC ranks agreement**: distribution of the agreement between compounds neighbors rankings. The lower the score, the lower the agreement among the compounds neighbors rankings
- **2D projection colored by CC ranks agreement**: *t*-SNE representation of the data, colored by their neighbor rank agreement
- **Outliers**: this represents the anomaly score for each signature. Signatures with scores above 0 are considered outliers
- **Key coverage**: the proportion of compounds shared across different CC space datasets. Any given compound may be present in all or some CC spaces (1–25, *x* axis)
- **2D projection colored by key coverage**: *t*-SNE representation of the data, colored by the compound coverage. Blue indicates higher compound presence across CC spaces
- **Cluster sizes**: the proportion of keys assigned to each cluster. We obtain the clusters by computing the distances between the signatures and then apply DBScan method to find the neighborhoods. The top 20 clusters are highlighted in the plot
- **Top clusters**: 2D representation of the top 20 clusters
- **MoA**: recapitulation of the NNs in the B1 space (MoA) using the current signature. This recapitulation is quantified by area under the receiver operating characteristic (AUROC) values, indicated in the figure title. Values >0.5 indicate random results
- **Anatomical Therapeutic Chemical (ATC)**: recapitulation of the NNs in the E1 space (therapeutic areas; ATC) using the current signature
- **Keys/features dimension**: the number of compounds and features across all CC spaces for the current signature. The values were normalized using $\log_{10}$. The current CC space dot is highlighted with black border and white background
- **CC wrt sign**: the number of keys of the current space that overlaps with the keys of all the other spaces colored according to the CC levels color legend. The numbers (ranging from 1 to 5) correspond to each space per level
- **Sign wrt CC**: the overlap among the keys of all the other spaces and the keys in the target space
- **Receiver operating characteristic (ROC) across CC**: recapitulation of the NNs of all CC spaces. Area under the ROC curve values summarize the performance. The area under the curve (AUC) values are sorted in descending order (higher values better recapitulation) and colored according to the CC levels color legend. The numbers (ranging from 1 to 5) correspond to each space per level

Generating diagnosis plots is a straightforward task with the CC package (Step 24). Further information on diagnosis plots and additional data analysis (for example, recapitulation of MoA and ATC) are found in our Gitlab. More importantly, Supplementary Table 2 provides specific guidance on how to interpret the results for each panel of the diagnosis plots.

Comparison with other methods

Chemical descriptors are at the core of cheminformatics. Indeed, there are many strategies aimed at encapsulating the structure of drug-like compounds in a format amenable for computational applications. **Extended connectivity fingerprints** represent the most popular approach to do so owing to their high interpretability, low computational cost and strong performance in various prediction tasks¹⁶. Other classical and popular representations include **substructure-based fingerprints**¹⁷, **Murcko scaffolds**¹⁸ and, more recently, **SMILES- and graph-based embeddings** built upon transformer architectures (for example, refs. 19,20). However, as stated in the 'Introduction', **bioactivity descriptors offer an alternative (and arguably more**

clinically relevant) perspective to navigate the chemical space in the search for new bioactive compounds with desired pharmacological properties. To our knowledge, there is no alternative strategy for processing and integrating user-provided biological data in response to chemical perturbations to generate customized bioactivity descriptors.

Experimental design

In this study, we illustrate the use of the protocol to build bioactivity spaces and signatures associated to small molecules in four distinct tasks, organized in increasing level of conceptual and technical complexity (Fig. 2):

1. Extend a preexisting CC space with additional bioactivity data: B1.002 space
2. Rebuild a preexisting CC space with a different strategy of processing the raw data: D1.002 space

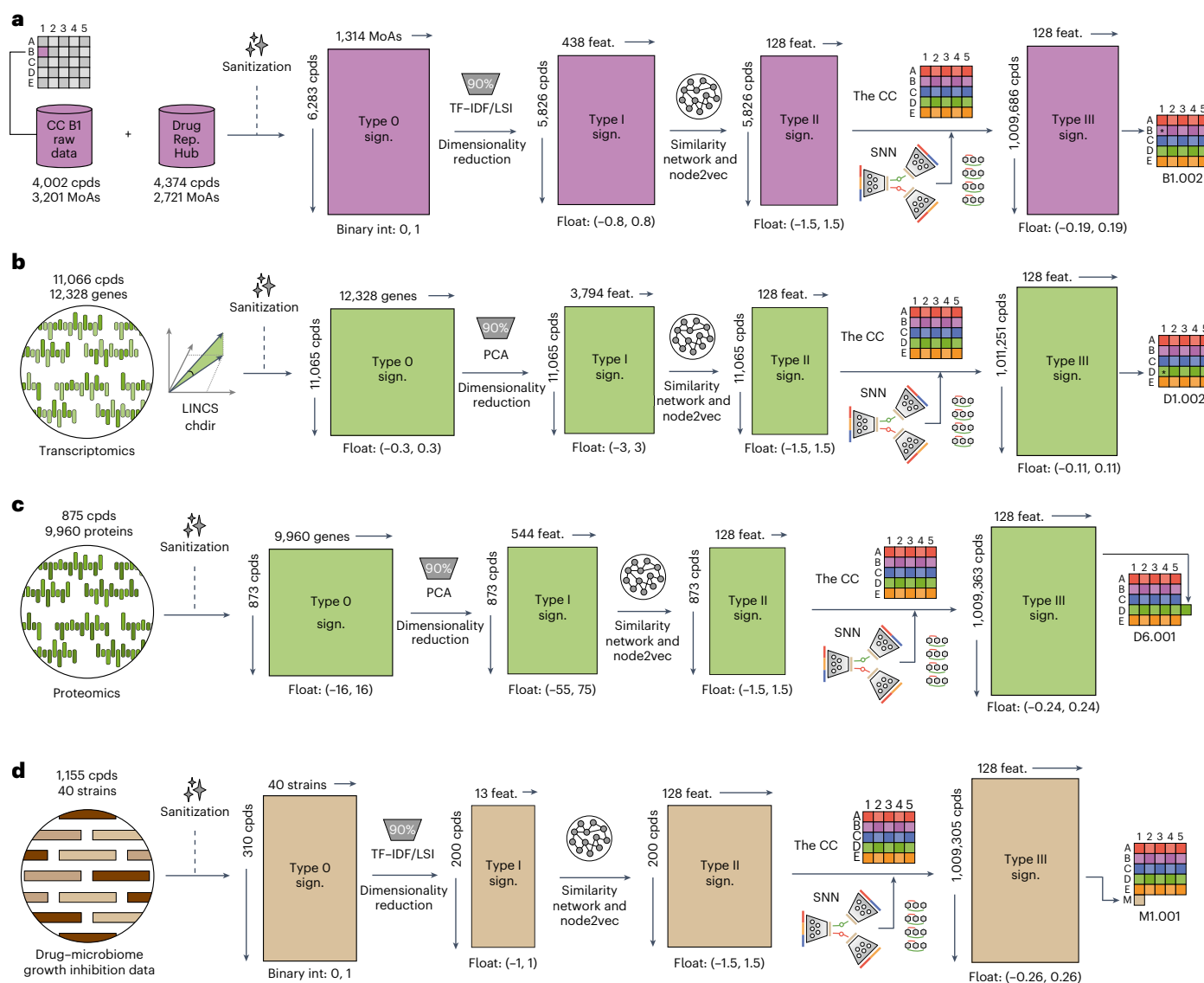


Fig. 2 | CC Protocols case examples. All steps along the pipeline are detailed in the procedure and the results are explained and discussed in the 'Anticipated results' section. **a**, Extending a pre-existing CC space with additional bioactivity data ('Anticipated results: Task 1'). Cpds, compounds; Drug Rep. Hub, Drug Repurposing Hub; feat., features; sign., signature. **b**, Rebuilding a pre-existing CC space with a different strategy of processing the raw data ('Anticipated results: Task 2').

results: Task 2') **c**, Creating a new CC space from a novel data type that fits the current CC universe (that is, human-derived data) ('Anticipated results: Task 3'). **d**, Creating a new CC space from a novel data type that does not fit the current CC universe (that is, measuring compound effects on bacterial species) ('Anticipated results: Task 4').

3. Create a new CC space from a novel data type that fits the current CC universe (that is, human-derived data): D6.001 space
4. Create a new CC space from a novel data type that does not fit the current CC universe (that is, measuring compound effects on bacterial species): M1.001 space

All four tasks are fully presented in the ‘Anticipated results’ section. Overall, in all cases, novel bioactivity data are processed, sanitized, compressed, harmonized and integrated in the CC universe of small-molecule signatures. Moreover, bioactivity signatures enable the exploration and visualization of large chemical libraries in a biologically relevant manner, the prediction of compound properties such as toxicity or the ability to revert gene expression (GEx) profiles associated with certain disease states, among others (Fig. 1c). The CC nomenclature to define the different bioactivity spaces first notes the space (that is, A1-*n*, B1-*n* and so on) and then numbers the dataset or processing protocols used (that is, 001, 002 and so on). Specific details on how to interpret the results in each bioactivity space and signatures (that is, diagnosis plots) are provided in the section ‘Overview of the CC data integration pipeline. Interpreting the results: diagnosis plots’ above.

Expertise needed to implement the protocol

The protocol can be implemented by anyone with basic experience with the Bash command line (mainly for the local installation of the CC) and basic scripting knowledge of the Python coding language (to actually run the designed protocol). Additionally, familiarity with high-performance computing (HPC) cluster and queuing systems is recommended in case of (1) handling of large amounts of data or (2) running the last integration step of the pipeline, for which parallelization or GPU computing is usually helpful (Step 19 and the ‘Limitations’ section).

Limitations

The CC data integration pipeline has some drawbacks and limitations that need to be acknowledged before use. First, as all frameworks designed to process external user-provided data, the integrity of the final outcome is ultimately tied to the quality of the input data: suboptimal, noisy or otherwise compromised data can substantially degrade the results and produce unreliable signatures. To help the user evaluate their signatures, the diagnosis plot provides a quantification of biological and chemical signals recapitulated by the newly generated spaces, intended to provide insights on the biomedical information captured by the user data.

It is also important to consider that input datasets having a small overlap with the CC compound universe may result in type III signatures mostly based on the chemical properties of the small molecules, since the lack of additional bioactivity data might impact SNN training. In line with this, it is likely that small molecules with no reported bioactivity in a specific CC space are eventually characterized by the so-called null signatures, corresponding to the average signature derived from the training set. To partially control this effect and handle additional problematic issues (see ‘Introduction. Type III signatures’), every inferred CC type III signature is accompanied by an applicability score that measures the overall degree of confidence in the compound signature. In this way, such applicability scores make it possible to filter out compounds that have unreliable signatures, potentially leading to more accurate and trustworthy results (see the ‘Overview of the CC data integration pipeline. Type III signatures’ section above).

Another limitation of most CC bioactivity signatures, similar to many machine and deep learning applications^{21,22}, is the lack of direct interpretability. Only type 0 signatures are directly explainable, since signature elements (columns) correspond to well-defined biological features (for example, chemical substructures, binding targets, clinical side effects and so on). However, type I, II and III signatures are abstract molecular representations for which experimental data (biological features) cannot be deconvoluted. In any case, we have already shown that CC signatures are useful across various tasks, and it is indeed possible to achieve a nuanced degree of interpretability by identifying which bioactivity signatures are more valuable and relevant for the predicted outcomes¹⁰.

Finally, another potential limitation of the presented pipeline is the relatively high computational cost associated with the generation of type III signatures. This step integrates all

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available bioactivity signatures in the CC and benefits from memory availability and extensive parallelization and GPU computing. We have designed a twin version of the type III signature fit method to allow the execution of this step in a predefined HPC cluster (Step 19). It is worth mentioning that computational cost is usually not a problem in the other steps of the protocol, as illustrated in the ‘Anticipated results’ section for various distinct input datasets. In addition, sufficient free hard disk space is essential for successfully downloading CC data and running the computational protocol smoothly. Overall, we encourage users to free +150 GB of their machines (see the ‘Troubleshooting’ section).

Materials

Equipment setup

Hardware

Workstation with 4 central processing units (CPUs) and 16 GB of random-access memory (RAM). Optionally, a computational cluster can be used to speed up the ‘signaturization’ process (mainly of type III signatures), exploiting wider task parallelization and GPU computing, as well as higher memory availability in case of inputting a large dataset to the CC integration pipeline.

Software

Linux or Linux-emulating operative systems. The CC package needs to be installed following the steps detailed in Steps 1–4. A Singularity image is built during the installation process, including all required software packages and their dependencies to run the CC pipeline in a safe environment. For other systems, the creation of the Singularity image might not be supported, and it might require a virtual environment. In addition, Apple computers with M processors (for example, M1, M2 or M3) may face unanticipated issues.

Other

To generate type III signatures, the user needs to get all type II signatures as well as additional dependencies from the current CC version. Signatures can be downloaded as plain data files directly from <https://chemicalchecker.com/downloads/root>, using the Python code we provide to do so (check Step 3), or accessing through the CC REST API (<https://chemicalchecker.com/help>, bottom of the page).

Data

In the ‘Anticipated results’ section, we describe results from four different tasks, encompassing distinct scenarios involving the processing of bioactivity data.

- In the first task (that is, extending a preexisting CC space with additional bioactivity data), we gathered publicly available data from the Drug Repurposing Hub²³
- In the second task (that is, rebuilding a preexisting CC space with a different strategy of processing the raw data), we used the same data as in our original publication⁹
- In the third task (that is, creating a new CC space from a novel data type that fits the current CC universe), we collected available proteomics data from the screening developed by Mitchell and co-workers¹⁴
- In the fourth and last task (that is, creating a new CC space from a novel data type that does not fit the current CC universe), we used the antibiotic data publicly released in the screening performed by Maier and co-workers²⁴

Procedure

▲ **CRITICAL** Steps 1–10 are preliminary. These are performed once. The main experiment starts at Step 11. Steps 11–24 all need to be performed for each experiment.

Stage 1: setting up the CC environment

● ~30 min

▲ **CRITICAL** Install the CC in a local computer as described in https://gitlab.bnb.irbbarcelona.org/packages/chemical_checker. The main steps are described below.

In short, we need to

- install Singularity (version>3.6)
- clone the CC repository to a local directory
- run the setup script (sudo permissions are needed)

After that, we need to

- adapt the CC configuration file to fit the user's CC environment
- clone the CC protocols repository to implement the work described in this article (see the 'Anticipated results' section)

1. Install Singularity

◆ TROUBLESHOOTING

```
$ sudo apt-get update && sudo apt-get install -y \
  build-essential \
  uuid-dev \
  libgpgme-dev \
  squashfs-tools \
  libseccomp-dev \
  wget \
  pkg-config \
  git \
  cryptsetup-bin \
  golang-go
$ export Version=3.8.0 && # adjust this as necessary \
wget https://github.com/sylabs/singularity/releases/download/
v${VERSION}/singularity-ce-${VERSION}.tar.gz && \
tar -xzf singularity-ce-${VERSION}.tar.gz && \
cd singularity-ce-${VERSION}
$ ./mconfig && \
make -C ./builddir && \
sudo make -C ./builddir install
```

2. Clone the CC repository to a local directory named PATH_TO_CC

◆ TROUBLESHOOTING

```
mkdir -p <PATH_TO_CC> && cd <PATH_TO_CC>
git clone http://gitlab.bnb.irbbarcelona.org/packages/chemical_checker.git
```

3. Run the CC setup script

◆ TROUBLESHOOTING

```
cd chemical_checker && sh setup/setup_chemicalchecker.sh
```

4. Manually edit the CC configuration file located at PATH_TO_CC/chemical_checker/setup/cc_config.json, referred to as PATH_TO_CC_CONFIG. In this file, the user must specify the variables CC_REPO, CC_TMP and SINGULARITY_IMAGE.

◆ TROUBLESHOOTING

- SINGULARITY_IMAGE should point to the CC singularity image created when running the CC setup script (see Step 3), usually located in \$HOME/chemical_checker/cc.simg, with \$HOME being the user's home directory

- CC_REPO should point to the chemical_checker directory inside PATH_TO_CC (that is, PATH_TO_CC/chemical_checker, see Steps 2 and 3)
 - CC_TMP should point to a local directory, defined and created by the user, where CC outcome files will be stored. This directory is referred to as PATH_TO_CC_TMP
5. Clone the CC protocols repository to a local directory referred to as PATH_TO_CC_PROTOCOLS. This repository includes all the notebooks necessary to implement the analyses presented in this manuscript.

```
mkdir -p <PATH_TO_CC_PROTOCOLS> && cd <PATH_TO_CC_PROTOCOLS>
git clone http://gitlab.bnb.irbbarcelona.org/packages/protocols.git
```

In brief, the repository contains four Python notebooks corresponding to the four tasks implemented herein (see the ‘Anticipated results’ section), along with the raw bioactivity data required to build the new CC spaces. The following points in this section provide an overview of the individual steps needed to generate new CC spaces and signatures.

◆ TROUBLESHOOTING

Stage 2: running a Jupyter Notebook with the CC image

● <1 min

6. Run a Jupyter Notebook in the CC protocols directory within the CC image by executing the following command (which is an alias to run the setup/run_chemicalchecker.sh file from the cloned CC repository).

◆ TROUBLESHOOTING

```
source ~/.bashrc
cd <PATH_TO_CC_PROTOCOLS>
chemcheck -d <PATH_TO_CC>/chemical_checker/package -c <PATH_TO_CC_CONFIG>
```

All notebooks needed to build the four spaces described in this manuscript are found in PATH_TO_CC_PROTOCOLS/protocols/notebooks.

7. Point the environment variable CC_CONFIG to the local CC configuration file (see Step 4).
8. Import the CC package itself as well as all necessary external libraries to run the analyses (for example, numpy, see examples in the ‘Anticipated results’ section and Supplementary Information).
9. Specify the level of log outputs printed along the procedure. The following piece of code refers to Steps 7, 8 and 9 and is already included in all the exemplary notebooks.

```
import os
os.environ['CC_CONFIG'] = '/aloy/home/acomajuncosa/cc_config.json'
# Modify path at will
from chemicalchecker import ChemicalChecker
ChemicalChecker.set_verbosity('DEBUG') # CRITICAL, ERROR, WARN, INFO
or DEBUG
```

Stage 3: gathering CC data

● ~45 min

▲ **CRITICAL** To successfully implement all CC functionalities illustrated in this work, the user needs to gather the CC bioactivity data from our web servers (for example, type II signatures, see Step 18). To illustrate the downloading process, we provide exemplary code that programmatically downloads the CC signatures, although the user can download individual files containing bioactivity signatures manually (<https://chemicalchecker.org/downloads/signatureX>, X being 0, 1, 2 or 3).

10. Download the data in a compressed tar.gz format and then uncompress the data in the directory PATH_TO_DATA, as shown in the notebook Download_Data.ipynb included in the CC protocols repository (see code below). Since the CC is subject to yearly updates, the size of the downloaded files may increase, as does the amount of publicly released bioactivity data. As of October 2024, the overall process takes 15 min to download and 30 min to decompress, requiring ~65 GB of free disk space and a stable wired internet connection (for example, Ethernet). For further details on what specific CC data are needed to implement each CC functionality, please see the Supplementary Information.

◆ TROUBLESHOOTING

```
import tarfile
import wget
import os
# Modify PATH at will
PATH_TO_OUTPUT = "/aloy/home/acomajuncosa/CC_DATA"
# Create path
os.makedirs(PATH_TO_OUTPUT, exist_ok=True)
# Download data
def download_data(PATH_TO_OUTPUT):
    os.chdir(PATH_TO_OUTPUT)
    wget.download("https://chemicalchecker.com/api/db/getFile/root/sign_links.tar.gz/", out=PATH_TO_OUTPUT)
download_data(PATH_TO_OUTPUT)
# Uncompress data
def decompress_data(PATH_TO_FILE, PATH_TO_DATA):
    os.makedirs(PATH_TO_DATA, exist_ok=True)
    with tarfile.open(PATH_TO_FILE, "r:gz") as tar:
        tar.extractall(path=PATH_TO_DATA)
# Modify PATHS at will
PATH_TO_FILE = "/aloy/home/acomajuncosa/CC_DATA/sign_links.tar.gz"
PATH_TO_DATA = "/aloy/home/acomajuncosa/CC_DATA/DATA"
decompress_data(PATH_TO_FILE, PATH_TO_DATA)
```

Main steps

▲ **CRITICAL** While preliminary steps are meant to be executed only once, the main steps of the protocol need to be run for each newly created CC space. The specific code developed to perform each individual task (see the ‘Experimental design’ section) is included in the CC protocols repository (see Step 5). As an example, the code shown in this section corresponds to the new M1.001 space (that is, task 4. The code for all the other tasks is available in the Supplementary Information). The overall timing of the main steps fully depends on the type and size of input data. For the sake of completeness, we specify the timing of all main steps for the four distinct tasks in the ‘Timing’ section.

Stage 4: creating a CC instance

● <1 min

11. Create a new CC instance by choosing a directory to allocate the results (local_cc_dir). Additionally, specify the location of existing CC data (PATH_TO_DATA, see Step 10).

◆ TROUBLESHOOTING

```
local_cc_dir = '../local_CC_M1' # Modify at will
PATH_TO_DATA = "/aloy/home/acomajuncosa/CC_DATA/DATA/" # Modify at will
cc_local = ChemicalChecker(local_cc_dir, dbconnect=False, custom_data_path=PATH_TO_DATA)
```

The typical CC folder architecture will be created inside `local_cc_dir`. The full directory will contain the computed signatures (from 0 to III) for the corresponding small molecules, while the reference set will include a nonredundant subset of the data, computed using the distance matrix among all compounds.

Stage 5: loading raw data

● <1 min, depending on the task (see the 'TIMING' section).

12. Check that the user input data conforms to the expected requirements. Essentially, the data must consist in a well-defined bioactivity matrix (for example, pandas data frame) containing N compounds (rows, indices) and M biological features (columns). For all the examples presented in this manuscript, we also provide the code to download the starting data from our online repository and store it in the corresponding data directory (not shown here) to be then loaded (see Step 13).
13. To start the signaturization process, load a preprocessed version of the raw bioactivity data (see Step 12). Depending on the format of the data, the loading process may differ substantially. In the 'Anticipated results' section and Supplementary Code, we have included examples in which the data is located in a CSV file and in a H5 file (see the code below for loading the input csv data file for the new M1.001 space). Note that the code needs to be adapted when using user-defined data.

```
inputFile="../data/M1/microbiota_raw.csv" # Modify at will
df=pd.read_csv(inputFile,index_col=0)
```

▲ **CRITICAL STEP** If bioactivity data are not properly loaded, the signaturization process will lead to unconfident results.

Stage 6: generating type 0 signatures

● 1–5 min depending on the task (see the 'TIMING' section for more explanation)

14. Once the bioactivity data is loaded, define a dataset name in the CC naming format, specifying the CC space the data will be dumped in (for example, M1.001). The CC nomenclature to define the different bioactivity spaces first notes the space (that is, A1- n , B1- n and so on) and then numbers the dataset or processing protocols used (that is, 001, 002 and so on).
15. Generate type 0 signatures. For further information about the generation of type 0 signatures, please see the 'Overview of the CC data integration pipeline. Type 0 signatures' section). The following code block refers to Steps 14 and 15.

◆ TROUBLESHOOTING

```
dataset = 'M1.001' # for example, B1.002, D1.002, D6.001, M1.001
sign0 = cc_local.signature(dataset, 'sign0')
sign0.clear_all()
sign0.fit(X=df.values, keys=list(df.index), features=list(df.columns))
```

Stage 7: generating type I signatures

● 10–30 min depending on the task (see the 'TIMING' section for more explanation)

16. Generate type I signatures using the code below. The code is straightforward and analogous to the previous step. The code also includes instructions to precompute compound NNs at type I signature level. For further information about the generation of type I signatures and the calculation of neighbors, please see the 'Overview of the CC data integration pipeline. Type I signatures and Type II signatures' sections).

◆ TROUBLESHOOTING

```
sign0 = cc_local.signature(dataset, 'sign0')
sign1 = cc_local.signature(dataset, 'sign1')
```

```
sign1.clear_all()
sign1.fit(sign0)
sign1 = cc_local.signature(dataset, 'sign1')
neig1 = cc_local.get_signature("neig1", "full", dataset)
neig1.clear_all()
neig1.fit(sign1)
```

Stage 8: generating type II signatures

● 1–10 min depending on the task (see the 'TIMING' section)

17. Generate type II signatures using the code below. As seen in previous steps, the code is straightforward and follows the same architecture as before. For further information about the generation of type II signatures, please see the section 'Overview of the CC data integration pipeline. Type II signatures'.

◆ TROUBLESHOOTING

```
sign1 = cc_local.get_signature('sign1', 'full', dataset)
neig1 = cc_local.get_signature('neig1', 'full', dataset)
sign2 = cc_local.signature(dataset, 'sign2')
sign2.clear_all()
sign2.fit(sign1, neig1, oos_predictor=False)
```

Stage 9: generating type III signatures

● 4–10 h depending on the task and the computational infrastructure (see the 'TIMING' section)

▲ **CRITICAL** The last stage in the signaturization process is the calculation of type III signatures.

18. Collect all type II signatures from the existing CC spaces (already downloaded in Step 10) along with type I and type II signatures for the extended or newly created space.
19. Decide where the type III signature calculation will be performed. Since the calculation of type III signatures is computationally expensive (see the 'Limitations' section), we run it on an HPC cluster to enable CPU parallelization or GPU computing. Indeed, we encourage users to run the generation of type III signatures in their own HPC cluster. Otherwise, it can be run locally in the same manner that previous steps, although the overall calculation may take longer to complete.
20. Additionally, when creating a new CC space, check the overlap between the input dataset and the CC universe of small molecules.
▲ **CRITICAL STEP** It is important to do this, because poor overlaps between both sets may eventually result in unreliable type III signatures (see the 'Limitations' section).
21. (optional) Set the complete_universe parameter.
▲ **CRITICAL STEP** Chemical signatures (A1, A3, A4 and A5) are automatically calculated for all those compounds that are not already listed within the CC universe (complete_universe='fast'). If the user also wants to include A2 signatures, a particularly slow process due to the generation of 3D conformers, the complete_universe parameter needs to be set to 'full'.
22. (optional) Provide the corresponding mappings between the input InChIKeys and the molecule InChIs in a Python dictionary. If this is not done, the pipeline will query online repositories (for example, PubChem) to derive the mappings, which may notably slow down the entire process.
23. Generate type III signatures. The code provided below refers to Steps 18–23 and, as all the code shown in the 'Procedure' section, corresponds to the case example from 'Anticipated results: Task 4 (M1.001)' and needs to be adapted to every particular exercise. For additional examples, please see Supplementary Code: Tasks 1–3.

◆ TROUBLESHOOTING


```
# CHECK OVERLAP
# Dataset Name
dataset = 'M1.001'
# Get CC universe
cc_universe = []
for dat in cc_local.datasets:
    if dat != dataset and dat.endswith('001'):
        cc_universe.extend(cc_local.get_signature('sign2', 'full', dat).keys)
cc_universe = set(cc_universe)
# Get sign2
sign2 = cc_local.signature(dataset, 'sign2')
# Get M1 molecules
m1_molecules = set(sign2.keys)
print("Number of molecules in the CC universe: " + str(len(cc_universe)))
# 1009293
print("Number of molecules in M1 sign2: " + str(len(m1_molecules))) # 200
print("Intersection CC & M1: " + str(len(cc_universe.intersection(m1_
molecules)))) # 188
# GET DATA
# Instantiation of sign3
sign3 = cc_local.signature(dataset, 'sign3')
sign3.clear_all()
# Create a list of sign2 to feed sign3 -- using the 25 CC spaces & M1
sign2_list = list()
# For each CC space
for ds in cc_local.coordinates:
    ds += '.001'
    sign2_list.append(cc_local.get_signature('sign2', 'full', ds))
# Append the new M1 space
sign2_list.append(cc_local.get_signature('sign2','full', dataset))
# In total, we now have 26 spaces
print(len(sign2_list)) # 26
# Get M1 sign1
sign1_self = cc_local.signature(dataset, 'sign1')
# Get M1 sign2
sign2_self = cc_local.signature(dataset, 'sign2')
# FIT SIGN3
mapp = None
"""
# Alternatively, you can provide your in-house python dictionary to map
InchiKey's to InChI's (see example below)
mapp = {
'LPXQRXLUHJKZIE-UHFFFAOYSA-N': 'InChI=1S/C4H4N6O/c5-4-6-2-1(3(11)7-4)8-10-
9-2/h(H4,5,6,7,8,9,10,11)',
'BZKPWHYZMXOIDC-UHFFFAOYSA-N': 'InChI=1S/C4H6N4O3S2/c1-2(9)6-3-7-8-4(12-3)
13(5,10)11/h1H3,(H2,5,10,11)(H,6,7,9)',
'XZWYZXLIPXDOLR-UHFFFAOYSA-N': 'InChI=1S/C4H11N5/c1-9(2)4(7)8-3(5)6/h1-2H3,
(H5,5,6,7,8)'
}
"""
```

▲ **CAUTION** COMPUTATIONALLY DEMANDING STEP - Consider running it in an HPC cluster

```
sign3.fit(sign2_list, sign2_self, sign1_self, sign2_universe=None,
complete_universe="fast", sign2_coverage=None, dbconnect=False,
mapping_dict=mapp)
```

Stage 10: generating diagnosis plots

● 1–45 min depending on the task and the plot type (see the ‘TIMING’ section)

24. To visualize the flow of the data along the integration pipeline, generate diagnosis plots for all the newly obtained signatures (0, 1, 2 and 3) with the code provided below (replacing *X* by 0, 1, 2 or 3, depending on the signature type under analysis). For further information about the generation and interpretation of diagnosis plots, please see the section ‘Overview of the CC data integration pipeline. Interpreting the results: diagnosis plots’).

◆ TROUBLESHOOTING

```
sign2 = cc_local.signature(dataset, 'sign2')
diag2 = sign2.diagnosis(ref_cctype='sign0')
diag2.canvas(size='small', savefig=True, savefig_kwargs={'dpi': 300})
```

Troubleshooting

Troubleshooting advice can be found in Table 1.

Table 1 | Troubleshooting

Step	Problem	Cause	Solution
1	Singularity is not properly installed	User has no sudo permissions	Install Singularity with a user having sudo permissions
2	The CC repository is not cloned properly	No internet connection or network instability	Fix internet connection issues and/or try cloning again without further changes
3	The setup script fails when creating the CC Singularity Image	Insufficient space left	Free up local disk space (~20 GB are needed for this step)
3	The setup script fails due to issues related with the wheel python package	Python version is not compatible	Use Python >3.7 or wheel = 0.37
4	The CC configuration file is not found	Unsuccessful cloning procedure	Clone the repository again or create the configuration file manually using the Gitlab template
5	The CC protocols repository is not cloned properly	No internet connection or network instability	Fix internet connection issues and/or try cloning again without further changes
6	The chemcheck command fails	The cc_config file has incorrect paths	Correct the setup/cc_config.json file (paths to the CC local repository, Singularity image and so on)
6	The chemcheck command provides multiple Jupyter notebook links	External logging generated by Jupyter	The user should look for the following link: <a href="http://<IP-ADDRESS>:<PORT>/lab?token=<TOKEN>">http://<IP-ADDRESS>:<PORT>/lab?token=<TOKEN>
10	The CC website is not available	No internet connection	Fix internet connection issues
11	Issues when creating a CC instance	(1) local_cc_dir does not exist, (2) PATH_TO_DATA does not exist, (3) data not in PATH_TO_DATA	(1) create local_cc_dir. To create again the instance, you need to delete the local_cc_dir, it does not overwrite, (2) create PATH_TO_DATA, (3) download data from the CC website and locate it in the proper directory (see Step 3)
15	Issues when generating type 0 signatures (a)	Insufficient memory to process data	(1) Use workstation with more RAM memory, (2) use an HPC cluster, (3) decrease the chunk_size* parameter, (4) reduce the dimensions of the input dataset
15	Issues when generating type 0 signatures (b)	Not enough molecules present in the produced signature	Set parameter check_keys to False

Table 1 (continued) | Troubleshooting

Step	Problem	Cause	Solution
15	Issues when generating type 0 signatures (c)	Not enough features present in the produced signature	(1) Increase parameter max_features (default is 10k), (2) set parameter check_features to False
16	Issues when generating type I signatures (a)	Insufficient memory to process data	(1) Use workstation with more RAM memory, (2) use an HPC cluster, (3) decrease the chunk_size* parameter, (4) reduce the dimensions of the input dataset
16	Issues when generating type I signatures (b)	Not enough molecules present in the produced signature	Set parameter check_keys to False
17	Issues when generating type II signatures (a)	Insufficient memory to process data	(1) Use workstation with more RAM memory, (2) use an HPC cluster, (3) decrease the chunk_size* parameter, (4) reduce the dimensions of the input dataset
17	Issues when generating type II signatures (b)	Node2vec or snap are not found	Both tools are available inside the created singularity image and should be defined in the cc_config.json file accordingly: "TOOLS": { "node2vec_exec": "/opt/node2vec", "snap": "/opt" }
23	Issues when generating type III signatures (a)	Calculation process taking too long	(1) Set the complete_universe parameter to 'fast' or False, (2) increase the number of CPUs, (3) reduce the size of the input dataset
23	Issues when generating type III signatures (b)	Insufficient space left	Large files are created when generating type III signatures. Consider removing unnecessary h5 files in the models folder when the process is finished
24	Issues when generating diagnosis plots	Insufficient memory	(1) Use workstation with more RAM memory, (2) use an HPC cluster
All	Issues related with hard disk space	Insufficient hard disk space	Consider removing data accumulated during the pipeline that is not necessary anymore: (1) the sandbox folder created in Step 1 and located in \$HOME/chemical_checker, (2) the compressed file containing CC data downloaded in Step 3 and so on

*chunk_size: the chunk_size parameter specifies the number of rows (molecules) in each chunk to be iteratively processed for predicting and writing output signatures to the sign3.h5 file.

Timing

Preliminary steps:

Steps 1–5, Stage 1: setting up the CC environment: 30 min

Steps 6–9, Stage 2: running a Jupyter Notebook with the CC image: <1 min

Step 10, Stage 3: gathering CC data: 45 min

Main steps:

Task 1:

Step 11, Stage 4: creating a CC instance: <1 min

Steps 12–13, Stage 5: loading raw data: <1 min

Steps 14–15, Stage 6: generating type 0 signatures: 1 min

Step 16, Stage 7: generating type I signatures: 10 min

Step 17, Stage 8: generating type II signatures: 2 min

Steps 18–23, Stage 9: generating type III signatures: 8 h in a 32-core CPU machine, 7 h in a 12-core GPU machine

Step 24, Stage 10: generating diagnosis plots: 2 min (small) and 45 min (medium)

Task 2:

Step 11, Stage 4: creating a CC instance: <1 min

Steps 12–13, Stage 5: loading raw data: <1 min

Steps 14–15, Stage 6: generating type 0 signatures: 5 min

Protocol

Step 16, Stage 7: generating type I signatures: 30 min
Step 17, Stage 8: generating type II signatures: 10 min
Steps 18–23, Stage 9: generating type III signatures: 10 h in a 32-core CPU machine,
8 h in a 12-core GPU machine
Step 24, Stage 10: generating diagnosis plots: 2 min (small) and 45 min (medium)

Task 3:

Step 11, Stage 4: creating a CC instance: <1 min
Steps 12–13, Stage 5: loading raw data: <1 min
Steps 14–15, Stage 6: generating type 0 signatures: 1 min
Step 16, Stage 7: generating type I signatures: 10 min
Step 17, Stage 8: generating type II signatures: 1 min
Steps 18–23, Stage 9: generating type III signatures: 10 h in a 32-core CPU machine,
8 h in a 12-core GPU machine
Step 24, Stage 10: generating diagnosis plots: 2 min (small) and 45 min (medium)

Task 4:

Step 11, Stage 4: creating a CC instance: <1 min
Steps 12–13, Stage 5: loading raw data: <1 min
Steps 14–15, Stage 6: generating type 0 signatures: 1 min
Step 16, Stage 7: generating type I signatures: 10 min
Step 17, Stage 8: generating type II signatures: 1 min
Steps 18–23, Stage 9: generating type III signatures: 6 h in a 32-core CPU machine,
4 h in a 12-core GPU machine
Step 24, Stage 10: generating diagnosis plots: 2 min (small) and 45 min (medium)

Anticipated results

In this section, we showcase the execution of the CC data integration pipeline in four different scenarios, as briefly described in the ‘Experimental design’ section (Fig. 2). For each task, we first introduce the nature of the raw bioactivity data (Dataset description) and we then present and discuss the results obtained along the pipeline (Signaturization results). The implemented procedure is detailed in the main steps of the procedure, and the individual code adapted to each task is included in Supplementary Code as well as in the example notebooks located in our Gitlab.

Task 1: extending a pre-existing CC space with additional bioactivity data

A global picture of the pipeline with detailed steps, numbers and timings is shown in Fig. 2a. Diagnosis plots for all signatures are shown in Extended Data Fig. 1 and Fig. 4.

B1.002 type 0, I, II and III signatures are available at <https://zenodo.org/records/14000467>.

Dataset description

The Drug Repurposing Hub²³ is a centralized, comprehensive and curated collection of FDA-approved drugs, preclinical tool molecules and clinically failed compounds for which safety has already been proven. All compounds are annotated with their chemical structures together with information about their MoAs, approved disease indications and clinical trial status, if available.

The original CC B1 space (B1.001, MoA) was built using available data from ChEMBL (v.22) and DrugBank (v.4). The main purpose of this exercise is to extend the B1 space (namely B1.002) with the data from the Drug Repurposing Hub, accounting for 4,374 compounds and 2,721 mechanisms of action.

Signaturization results, ~8 h

First, we observed that the overlap between the original (ChEMBL and DrugBank) molecule–MoA pairs and those coming from the Drug Repurposing Hub was low, with only 3,342 out of

the 11,971 new pairs already annotated in the original CC B1.001 space, which shows the value of incorporating this dataset. We merged all data (6,944 compounds and 4,626 MoAs) and started the signaturization process. The 3,312 seldom occurring MoAs and 661 compounds exhibiting null bioactivity profiles were filtered out in the first processing step (min_feature_abs and min_keys_abs parameters, respectively). Eventually, we obtained type 0 signatures for 6,283 compounds with 1,314 mechanisms of action (Step 15). As illustrated in the diagnosis plots, the new B1.002 type 0 signatures were binary and highly redundant (Extended Data Fig. 1a). In the next step, we reduced data dimensionality and obtained type I signatures for 5,826 compounds, with 438 PCA components capturing 90% of the variance (Step 16). We then harmonized the number of features to match the CC data and generated type II signatures for these 5,826 molecules (128 features, Step 17). Finally, we integrated the new B1.002 space together with the rest of the CC spaces and generated type III signatures for all 1,009,686 compounds (5,826 and 1,009,293 in B1.002 and the CC universe, respectively, with an overlap of 5,433 compounds, Steps 18–23).

An important consideration is that similarities at the raw data level (type 0 signatures) may be partially diluted along the data contextualization pipeline (for example, at type III signature level), when the data is eventually compressed and integrated with all the bioactivity data already contained in the CC. To further illustrate which spaces may be more affected in this regard, we have designed an evaluation process in which, for each new CC space and signature type, we assess the ability to recapitulate NN from a signature type at a lower level (Fig. 3a).

- First, for the newly created B1.002 space, we defined a cutoff cosine distance for all signature types (0, I, II and III, that is, 0, I, II and III) at a *P* value of 0.01 (10,000 randomly selected compounds × 10 subsamples, considering the average value)
- Then, for each combination of signA–signB, we evaluated the ability of signB to recapitulate NN defined at signA level with a *P* value of 0.01 (for each combination, 2,500 random molecules, 5 subsamples; totaling 6 combinations of signature type pairs)

In the case of the B1.002 space, we observed that type III signatures were able to greatly recapitulate similarities observed at the raw bioactivity data level (type 0 signatures, AUROCs ~0.85; Fig. 3b). In addition, we also observed that applicability values associated with type III signatures were fairly in line with the values obtained for the original CC bioactivity spaces (Fig. 3c).

Interestingly, we observed that, apart from including a higher number of compounds and MoAs, the new B1.002 space was less redundant than B1.001 (Extended Data Fig. 1 and Fig. 4). A *t*-SNE 2D representation of these two spaces show how the incorporation of new data populates some areas of the target space not previously covered, and that the new B1.002 space was able to better recapitulate known MoAs and ATCs for annotated compounds (Extended Data Fig. 2).

Task 2: rebuilding a pre-existing CC space with a different strategy of processing the raw data

A global picture of the pipeline with detailed steps, numbers and timings is shown in Fig. 2b. Diagnosis plots for all signatures are shown in Extended Data Figs. 3 and 4.

D1.002 type 0, I, II and III signatures are available at <https://zenodo.org/records/14000494>.

Dataset description

GEx responses after compound perturbation provide a convenient way to connect diseases to drugs. Since the release of the Connectivity Map and LINCS resources, compound GEx signatures have allowed the identification of potential treatments for diverse diseases (for example, refs. 12,25,26) and have provided new insights on compounds' MoAs, target and sensitivity responses. Given the substantial value of these screenings, conscious efforts have been carried out to remove potential sources of noise and inherent biases, unearthing higher and more robust biological signals. Particularly relevant has been the implementation of the characteristic direction (chdir) method²⁷, which, having been tailored in the context of compound perturbation responses, is not only more suitable for detecting meaningful GEx changes, but also provides a means to identify non-robust, noisy profiles. Thus, we reprocessed raw LINCS data using the chdir GEx profile protocol, as indicated by the authors²⁷. Eventually, we ended up with 11,066 compounds (rows) and 12,328 genes (columns), which substantially enhanced the

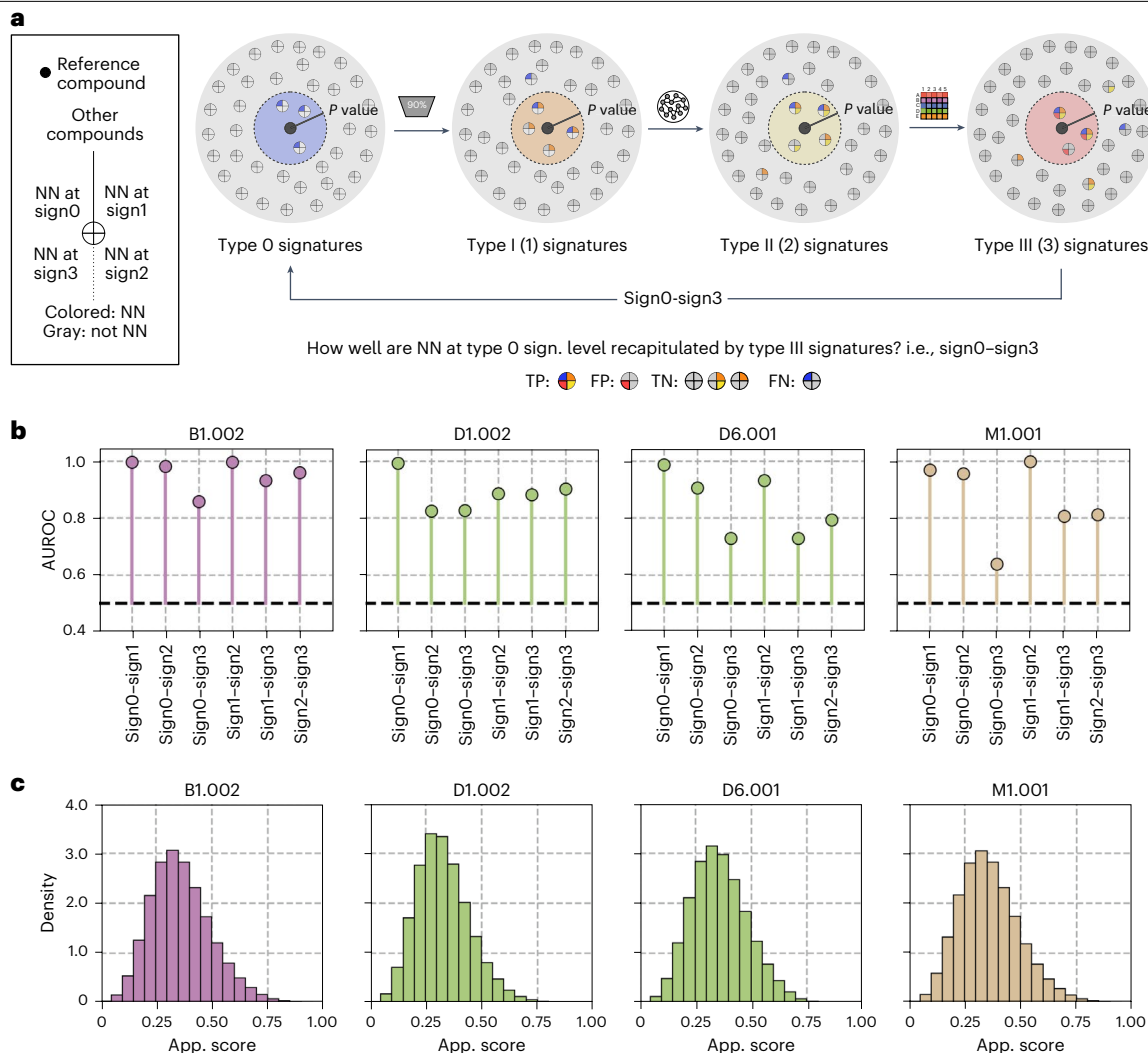


Fig. 3 | Quantitative evaluation of the signaturization process. a, A graphical representation of the validation strategy (from left to right). A reference compound (black circle) has a set of NN defined at type 0 signature level. Those NN may or may not be NN along the data compression, harmonization and integration pipeline. The better the ability to recapitulate NN defined at type 0 signature level by any other signature type, the fewer amount of information is lost along the computational pipeline. FN, false negatives; FP, false positives; TN, true negatives; TP, true positives. **b**, For each newly created CC space and pair of signature types (signA–signB), ability (y axis, AUROC) from signBs to recapitulate

NN defined at signA level. In brief, we defined a cutoff cosine distance for all signature types (0, I, II and III) at a *P* value of 0.01 (10,000 randomly selected compounds × 10 subsamples, considering the average value). Then, for each combination of signA–signB, we evaluated the ability of signB to recapitulate NN defined at signA level with a *P* value of 0.01 (for each combination, 2,500 random molecules, 5 subsamples; totalling 6 combinations of signature type pairs). **c**, For each newly created CC space, distribution of applicability values (app. score) associated to the generated type III signatures.

interpretability compared with the previous processing method. In the next paragraphs we exemplify the rebuilding of the D1 space with this new preprocessing protocol (D1.002).

Signaturization results, ~9 h

After processing the data, we started the rebuilding of the D1 space (D1.002) with 11,066 compounds (rows) and 12,328 genes (columns), while our old processing (D1.001) contained 14,098 compounds and 7,967 Connectivity Map features. This reduction in the number of compounds is due to the additional chdir filtering step that accounts for the robustness in the GEx replicates. Now, each compound–gene pair was associated with a numerical value representing the induced GEx change.

After creating the new CC instance and loading the preprocessed bioactivity data (Steps 11–13), we obtained type 0 signatures for 11,066 compounds against 12,328 genes (Step 15). It is important to note that, in the generation of type 0 signatures, we set the number

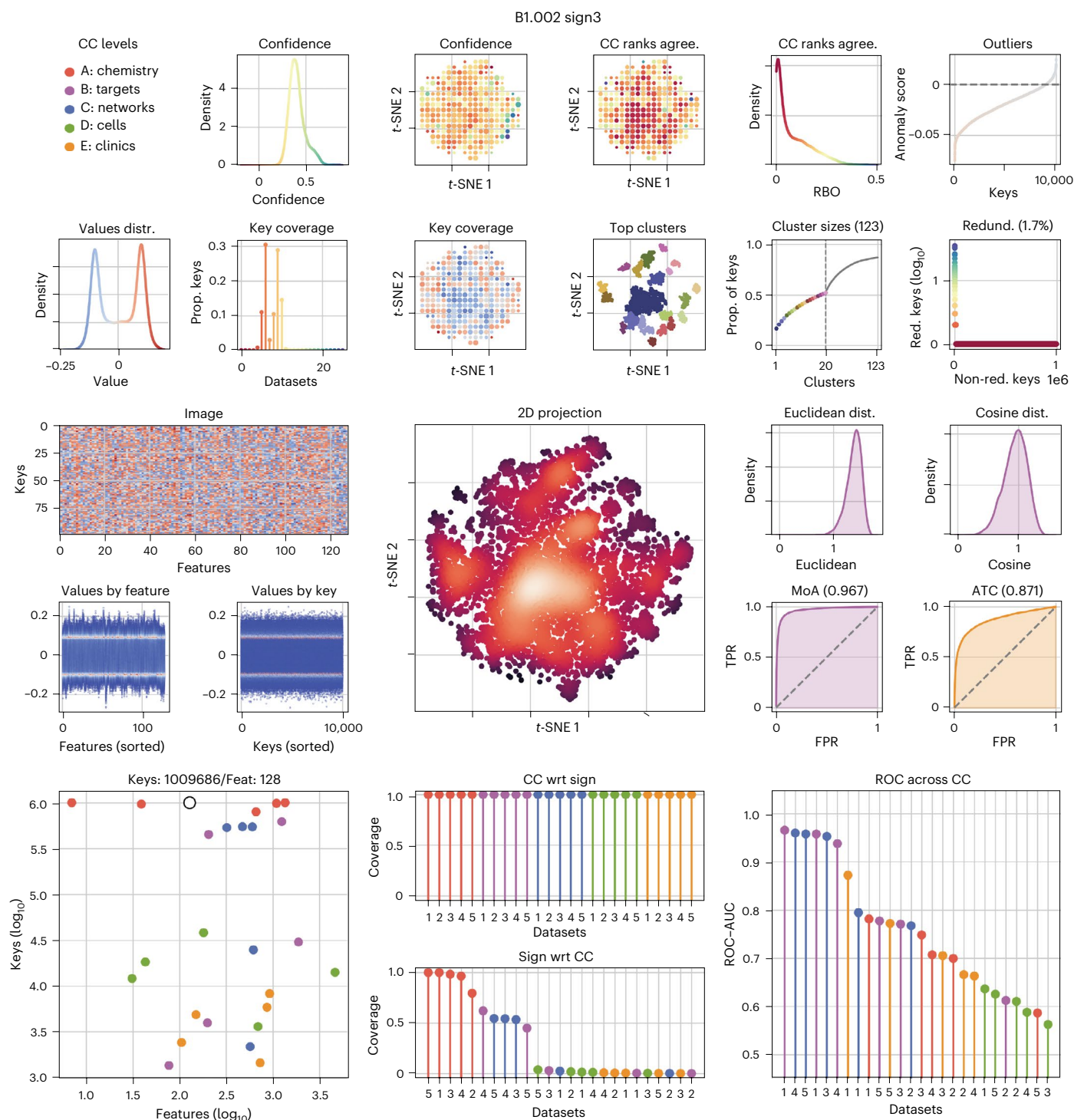


Fig. 4 | Extended diagnosis plots for B1.002 type III signatures (see ‘Anticipated results: Task 1’). For further information about diagnosis plots, please see the section ‘Overview of the CC data integration pipeline. Interpreting the results: diagnosis plots’ and check our Gitlab repository. Agree., agreement; dist., distances;

distr., distribution; FPR, false positive rate; prop., proportion; RBO, rank-biased overlap; red., redundant; redund., redundancy; TPR, true positive rate; wrt, with respect to.

of maximally considered features in the sanitization process up to 13,000, since the default value is 10,000 (please see Troubleshooting and Supplementary Code: Task 2). In addition, we observed that signatures were not redundant and that most values were in the $(-0.05, 0.05)$ range (Extended Data Fig. 3a).

Then, we obtained type I signatures for 11,065 compounds with 3,794 dimensions and generated harmonized type II signatures for these 11,065 compounds (Extended Data Fig. 3b,c and Steps 16 and 17, respectively).

Finally, we integrated the D1.002 space with the rest of CC spaces and obtained type III signatures for all 1,009,305 compounds (Extended Data Fig. 4 and Steps 18–23).

To assess the capacity of type I, II and III signatures to recapitulate NN defined at type 0 signature level (raw bioactivity data), we implemented the methodology described in the previous section ('Anticipated results: Task 1') and illustrated in Fig. 3a. In this task, we also observed that type III signatures were able to greatly recapitulate similarities defined at the raw bioactivity data level (type 0 signatures, AUROCs of ~ 0.83 ; Fig. 3b) and showed applicability values in the same range of preexisting CC bioactivity spaces (Fig. 3c).

A downstream comparison with the space obtained from our old processing (D1.001) confirmed the overall improved consistency of this new space, as illustrated by the better recapitulation of known drug bioactivity associations, including their MoA and therapeutic (ATC) areas (Extended Data Fig. 5). This exercise highlights the importance not only of the quality of the raw data employed to develop the bioactivity descriptors but also of the preprocessing strategies.

Task 3: creating a new CC space from a novel data type that fits the current CC universe

A global picture of the pipeline with detailed steps, numbers and timings is shown in Fig. 2c. Diagnosis plots for all signatures are shown in Extended Data Figs. 6 and 7.

D6.001 type 0, I, II and III signatures are available at <https://zenodo.org/records/14000554>.

Dataset description

Deconvoluting the MoAs of bioactive compounds within human cells is fundamental to tailor the eventual therapeutic effects they may exert. To study changes in protein levels induced by the presence of pharmacological agents, Mitchell et al.¹⁴ screened a collection of 875 small-molecule perturbagens in HCT116 cells and derived the so-called proteome fingerprints for them, accounting for the changes in protein abundance caused after 24 h of treatment and achieving deep proteome coverage ($>9,000$ quantified proteins). In short, they observed that over half of the quantified proteome changed by at least twofold with at least one compound, and 98% of the proteins showed a variation of five standard deviations produced by at least one compound.

Although cell-based assays are indeed included in the CC (for example, transcriptomics in D1 or sensitivity profiles in D2), proteomics data are not specifically considered. The main purpose of this exercise is to create a new CC space with the proteomics data presented above (namely D6.001) accounting for 875 compounds and 9,960 proteins.

Signaturization results, 9 h

After creating the new CC instance and loading the preprocessed bioactivity data (Steps 11–13), we found that 12.5% of the pairs did not have a reported ($\log_2$) fold change measurement (NaNs). We then obtained type 0 signatures for 873 compounds against 9,960 proteins (Step 15) and we observed that signatures were binary and completely nonredundant (Extended Data Fig. 6a). After that, we reduced their dimensionality by generating type I signatures for 873 compounds with 544 features (Step 16) and we then obtained type II signatures for these 873 compounds, at that point already harmonized with the other CC spaces (128 features, Step 17). Finally, we derived type III signatures for these 873 compounds together with all the CC universe of small molecules, totaling to 1,009,363 signatures (Extended Data Fig. 7 and Steps 18–23).

To assess the capacity of type I, II and III signatures to recapitulate NN defined at type 0 signature level (raw bioactivity data), we implemented the methodology described in the

section 'Anticipated results: Task 1' and illustrated in Fig. 3a. In brief, we observed that type III signatures were mostly able to recapitulate similarities defined at the raw bioactivity data level (type 0 signatures, AUROCs of ~0.73; Fig. 3b), although the performance was markedly lower than in other new spaces (that is, B1.002 and D1.002). In addition, we also observed that applicability values associated to type III signatures were in the same numerical range than in preexisting CC bioactivity spaces (Fig. 3c).

Additionally, when we compared the ability to recapitulate known MoA of both proteomics signatures with those derived from transcriptomics, observing that proteomics-based D6.001 type III signatures were able to identify known MoAs at the same level than the new D1.002 signatures, and better than the original D1.001 space (Extended Data Fig. 8a,c,e). Additionally, we also assessed the ability of D6.001 to capture therapeutic areas (ATC), finding that it recapitulates better known ATC annotations than the transcriptional spaces, perhaps reflecting protein abundance is a closer proxy for clinical outcomes (Extended Data Fig. 8b,d,f). We also asked whether proteomics-derived signatures were able to recapitulate NNs in transcriptomics signatures, finding a moderate capacity (AUROCs of 0.72 and 0.74 for D1.001 and D1.002, respectively), which indicates a high degree of complementarity between the two bioactivity spaces (Extended Data Fig. 8g,h).

Task 4: creating a new CC space from a novel data type that does not fit the current CC universe

A global picture of the pipeline with detailed steps, numbers and timings is shown in Fig. 2d.

Diagnosis plots for all signatures are shown in Extended Data Figs. 9 and 10.

M1.001 type 0, I, II and III signatures are available at <https://zenodo.org/records/14000572>.

Dataset description

Nonantibiotic marketed drugs have been recently associated with changes in the composition of the human gut microbiome²⁸. Such unforeseen activity may explain potential antibiotic-like side effects, might promote antibiotic resistance and may eventually contribute to a substantial decrease in microbiome diversity²⁴. To unravel potential relationships between nonantibiotic drugs and microbes, Maier et al.²⁴ screened a collection of drug-like compounds (including human-targeted drugs, antibiotics, anti-infective compounds and veterinary drugs, among others) against a set of 40 representative gut bacterial strains. Indeed, they observed that 24% of the drugs having human targets were able to inhibit the growth of at least one gut bacterial strain in vitro. We exhaustively processed their data and defined a binary growth inhibition matrix between tested compounds and strains. The novel bioactivity data involved 1,155 compounds (marketed drugs, rows) tested against 40 representative gut bacterial strains (columns). The main purpose of this exercise is to create a new CC space (that is, M1.001) where the chemical compounds exert their bioactivities on microbial species in the human microbiota.

Signaturization results, ~5 h

After creating the new CC instance and loading the preprocessed bioactivity data (Steps 11–13), we obtained type 0 signatures for 310 compounds against 40 gut bacterial strains (Step 15). In fact, we found that 769 molecules exhibited null bioactivity profiles and were thus filtered out from the signaturization process (min_keys_abs parameter). Additionally, 76 compounds were removed due to an excessive proportion of positive features (max_keys_freq parameter).

After that, we obtained type I signatures for 200 compounds against 13 features (Step 16), substantially reducing the number of dimensions of the data (Extended Data Fig. 9).

We then generated type II signatures for these molecules (Step 17), harmonizing the number of features to all the other spaces in the CC (128 features).

Finally, we derived type III signatures for these 200 compounds and for all the remaining molecules in the CC universe (1,009,305 molecules, in total, Steps 18–23).

To assess the capacity of type I, II and III signatures to recapitulate NN defined at type 0 signature level (raw bioactivity data), we implemented the methodology described in the section 'Anticipated results: Task 1' and further illustrated in Fig. 3a. Again, we observed

a decreased capacity for type III signatures to recapitulate similarities defined at the raw bioactivity data (type 0 signatures, AUROCs of ~0.64; Fig. 3b) compared with the three previous examples (that is, B1.002, D1.002 and D6.001). This lower performance highlighted the very orthogonal nature of the bioactivities in the M space (that is, effect of molecules on microbiota) and the rest of the CC, clearly dominated by human data. In addition, we also observed that applicability values associated to type III signatures were in the same numerical range than in preexisting CC bioactivity spaces (Fig. 3c).

The construction of the M1.001 space was challenging since, despite the apparent initially high number of compounds and features (that is, microbial species), the final number of compound–microbe relationships was quite limited. Nevertheless, we still observed a remarkable capacity of type III signatures to recapitulate known MoAs and ATC for annotated compounds (Extended Data Fig. 10).

Conclusions

Chemical signatures encode the physicochemical and structural properties of small molecules in the form of numerical descriptors, and they are at the core of chemical comparisons and search algorithms²⁹. Recently, the widespread availability of bioactivity data has led to enhanced compound representations that capture their biological effects along with their chemical structures. Yet such bioactivity descriptors are scarce, limited mostly to a small number of well-documented molecules. To overcome this problem, we implemented a collection of deep neural networks able to leverage the experimentally determined bioactivity data associated to small molecules and infer the missing bioactivity signatures for any compound of interest¹⁰. Originally, our strategy related to bioactivities of 25 different types (including target profiles, cellular responses and clinical outcomes), as described in the CC publication⁹. However, unlike typical chemical descriptors that remain static, our bioactivity signatures are dynamic, evolving as new bioactivity data accumulates in databases or new strategies to process it appear.

We have now made available the complete computational protocol to modify or generate novel bioactivity spaces and signatures, describing the main steps needed to leverage diverse bioactivity data with the CC using the predefined data curation pipeline. Moreover, we have illustrated the functioning of the protocol through four specific examples, including the incorporation of new compounds to an already existing bioactivity space, a change in the data preprocessing without altering the underlying experimental data and the creation of two novel bioactivity spaces from scratch.

Despite the regular updates, the expansion of the CC resource beyond the original 25 spaces is constrained by the availability and quality of public datasets. However, the systematic measurement of drug-induced perturbations in biological systems through omics technologies, together with advancements in phenotypic screening, are becoming increasingly common in the public and private sectors, providing an ever-growing corpus of biomolecular information. Besides, given the vast diversity of drug-like small molecules and the consolidation of generative artificial intelligence as a novel strategy to design new chemical entities, computational tools are progressively becoming essential for initially estimating the biological effects of compounds. Overall, we envision that this protocol to create new bioactivity spaces and the associated descriptors will serve as a pivotal tool for exploring the bioactivity spectrum of compounds, effectively bridging the existing knowledge gap between the chemical structure and the exerted biological effects of small molecules.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

Input data for all tasks presented in this manuscript are publicly available. Preprocessed bioactivity data are available in the CC Protocols Gitlab repository (<https://gitlab.bnb.irbbarcelona.org/packages/protocols>). Bioactivity signatures generated in this work are made available in the corresponding Zenodo repositories. B1.002 type 0, I, II and III signatures are available in

<https://zenodo.org/records/14000467>. D1.002 type 0, I, II and III signatures are available in <https://zenodo.org/records/14000494>. D6.001 type 0, I, II and III signatures are available in <https://zenodo.org/records/14000554>. M1.001 type 0, I, II and III signatures are available in <https://zenodo.org/records/14000572>.

Code availability

All code needed to implement the CC protocol is fully available in the CC Gitlab repository (https://gitlab.bnb.irbbarcelona.org/packages/chemical_checker). Additionally, all code needed to reproduce the exemplary tasks presented in this work is located in the CC Protocols Gitlab repository (<https://gitlab.bnb.irbbarcelona.org/packages/protocols>).

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References

- Kumar, A. & Zhang, K. Y. J. Advances in the development of shape similarity methods and their application in drug discovery. *Front. Chem.* **6**, 315 (2018).
- Polishchuk, P. G., Madzhidov, T. I. & Varnek, A. Estimation of the size of drug-like chemical space based on GDB-17 data. *J. Comput. Aided Mol. Des.* **27**, 675–679 (2013).
- Cereto-Massague, A. et al. Molecular fingerprint similarity search in virtual screening. *Methods* **71**, 58–63 (2015).
- Yang, J., Cai, Y., Zhao, K., Xie, H. & Chen, X. Concepts and applications of chemical fingerprint for hit and lead screening. *Drug Discov. Today* **27**, 103356 (2022).
- Holbeck, S. L., Collins, J. M. & Doroshow, J. H. Analysis of Food and Drug Administration-approved anticancer agents in the NCI60 panel of human tumor cell lines. *Mol. Cancer Ther.* **9**, 1451–1460 (2010).
- Seashore-Ludlow, B. et al. Harnessing connectivity in a large-scale small-molecule sensitivity dataset. *Cancer Discov.* **5**, 1210–1223 (2015).
- Campillos, M., Kuhn, M., Gavin, A. C., Jensen, L. J. & Bork, P. Drug target identification using side-effect similarity. *Science* **321**, 263–266 (2008).
- Petrone, P. M. et al. Rethinking molecular similarity: comparing compounds on the basis of biological activity. *ACS Chem. Biol.* **7**, 1399–1409 (2012).
- Duran-Frigola, M. et al. Extending the small-molecule similarity principle to all levels of biology with the Chemical Checker. *Nat. Biotechnol.* **38**, 1087–1096 (2020).
- Bertoni, M. et al. Bioactivity descriptors for uncharacterized chemical compounds. *Nat. Commun.* **12**, 3932 (2021).
- Comajuncosa-Creus, A., Lenes, A., Sanchez-Palomino, M., Dalton, D. & Aloy, P. Stereochemically-aware bioactivity descriptors for uncharacterized chemical compounds. *J. Cheminform.* **16**, 70 (2024).
- Pauls, E. et al. Identification and drug-induced reversion of molecular signatures of Alzheimer's disease onset and progression in App(NL-G-F), App(NL-F), and 3xTg-AD mouse models. *Genome Med.* **13**, 168 (2021).
- Anglada-Girotto, M. et al. Combining CRISPRi and metabolomics for functional annotation of compound libraries. *Nat. Chem. Biol.* **18**, 482–491 (2022).
- Mitchell, D. C. et al. A proteome-wide atlas of drug mechanism of action. *Nat. Biotechnol.* **41**, 845–857 (2023).
- Offensperger, F. et al. Large-scale chemoproteomics expedites ligand discovery and predicts ligand behavior in cells. *Science* **384**, eadk5864 (2024).
- Rogers, D. & Hahn, M. Extended-connectivity fingerprints. *J. Chem. Inf. Model.* **50**, 742–754 (2010).
- Durant, J. L., Leland, B. A., Henry, D. R. & Nourse, J. G. Reoptimization of MDL keys for use in drug discovery. *J. Chem. Inf. Comput. Sci.* **42**, 1273–1280 (2002).
- Bemis, G. W. & Murcko, M. A. The properties of known drugs. 1. Molecular frameworks. *J. Med. Chem.* **39**, 2887–2893 (1996).
- Shin, B., Park, S., Kang, K. & Ho, J. Self-attention based molecule representation for predicting drug-target interaction. Preprint at <https://doi.org/10.48550/arXiv.1908.06760> (2019).
- Zhou, G. et al. Uni-Mol: a universal 3D molecular representation learning framework. *ChemRxiv* <https://doi.org/10.26434/chemrxiv-2022-jjm0j> (2022).
- Jimenez-Luna, J., Grisoni, F., Weskamp, N. & Schneider, G. Artificial intelligence in drug discovery: recent advances and future perspectives. *Expert Opin. Drug Discov.* **16**, 949–959 (2021).
- Vamathevan, J. et al. Applications of machine learning in drug discovery and development. *Nat. Rev. Drug Discov.* **18**, 463–477 (2019).
- Corsello, S. M. et al. The Drug Repurposing Hub: a next-generation drug library and information resource. *Nat. Med.* **23**, 405–408 (2017).
- Maier, L. et al. Extensive impact of non-antibiotic drugs on human gut bacteria. *Nature* **555**, 623–628 (2018).
- Sawada, R., Iwata, M., Tabei, Y., Yamato, H. & Yamanishi, Y. Predicting inhibitory and activatory drug targets by chemically and genetically perturbed transcriptome signatures. *Sci. Rep.* **8**, 156 (2018).
- Chen, B. et al. Reversal of cancer gene expression correlates with drug efficacy and reveals therapeutic targets. *Nat. Commun.* **8**, 16022 (2017).
- Clark, N. R. et al. The characteristic direction: a geometrical approach to identify differentially expressed genes. *BMC Bioinform.* **15**, 79 (2014).
- Weersma, R. K., Zherakova, A. & Fu, J. Interaction between drugs and the gut microbiome. *Gut* **69**, 1510–1519 (2020).
- Fernandez-Torras, A., Comajuncosa-Creus, A., Duran-Frigola, M. & Aloy, P. Connecting chemistry and biology through molecular descriptors. *Curr. Opin. Chem. Biol.* **66**, 102090 (2022).

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Author contributions

M.B., M.D.-F., O.G.-P. and P.A. designed the study. M.B., M.D.-F., O.G.-P., M.L., Y.M. and N.S. implemented the computational pipeline. A.C.-C., A.F.-T., N.K., E.P.-L., G.R.-G. and E.V. created the different novel bioactivity spaces. A.C.-C. and P.A. wrote the manuscript. All authors analyzed the results, read and approved the manuscript.

Competing interests

The authors declare no conflict of interest.

Additional information

Extended data is available for this paper at <https://doi.org/10.1038/s41596-025-01167-3>.

Supplementary information The online version contains supplementary material available at <https://doi.org/10.1038/s41596-025-01167-3>.

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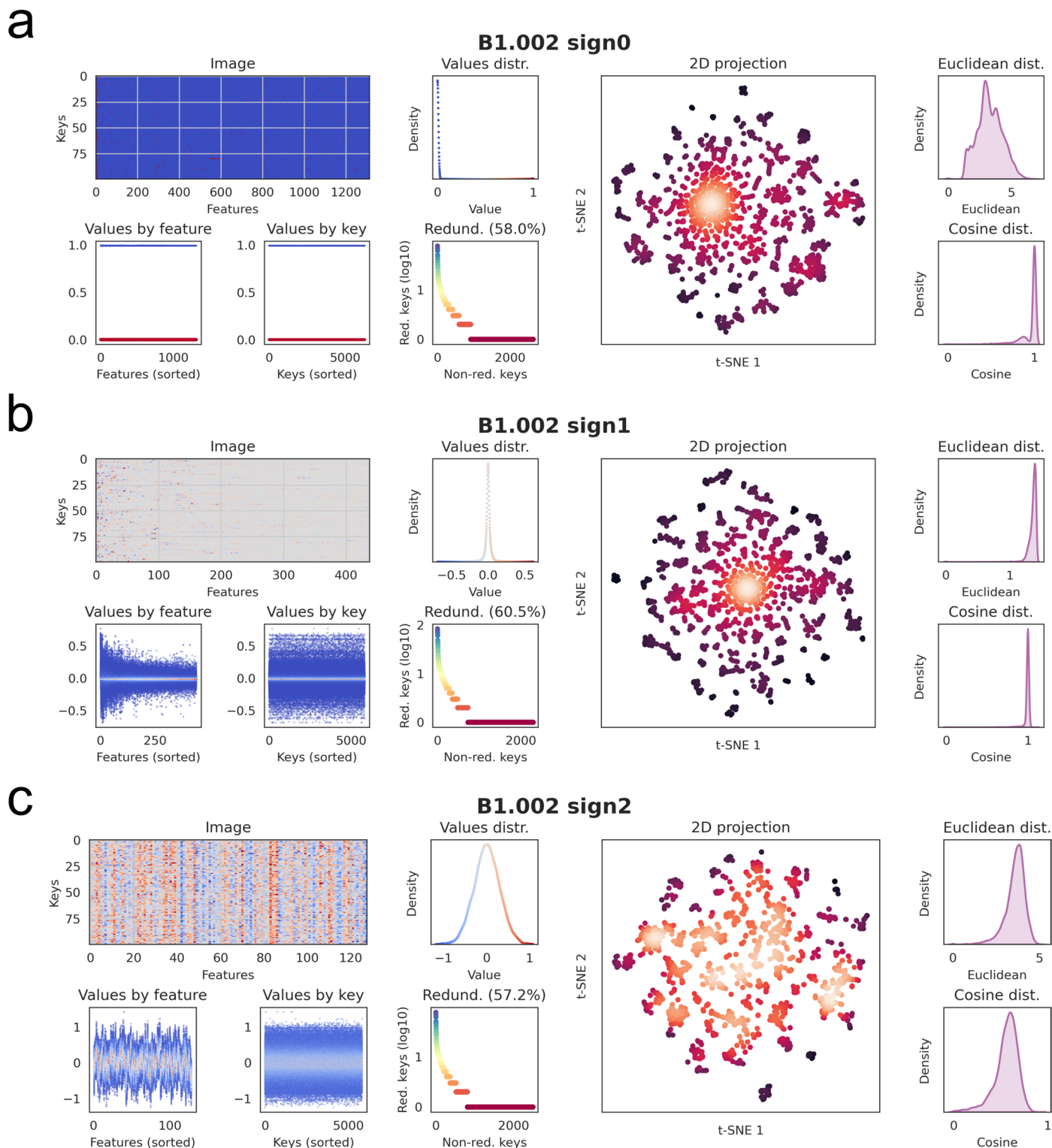
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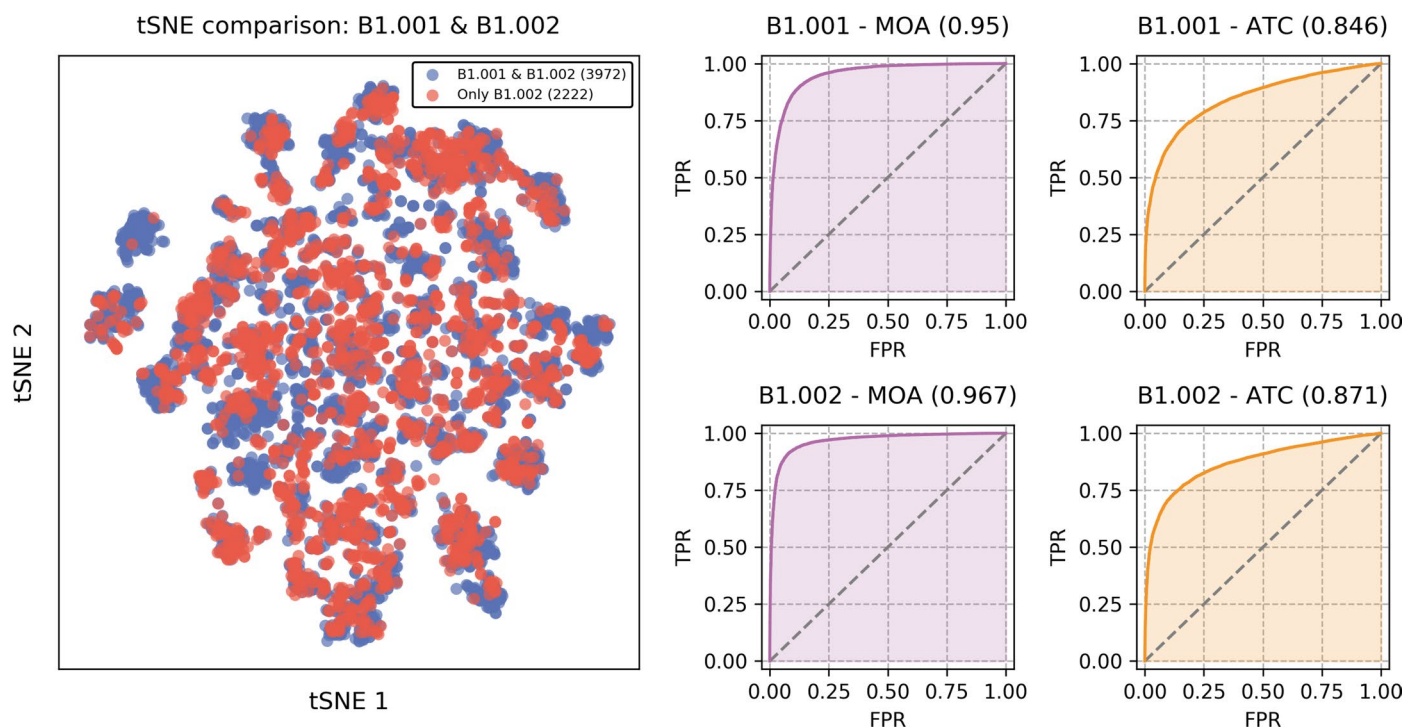
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Protocol



Extended Data Fig. 1 | Diagnosis plots for the B1.002 space (please see Anticipated Results: Task 1). **a)** type 0 signatures, **b)** type I signatures and **c)** type II signatures. For further information about diagnosis plots, please see

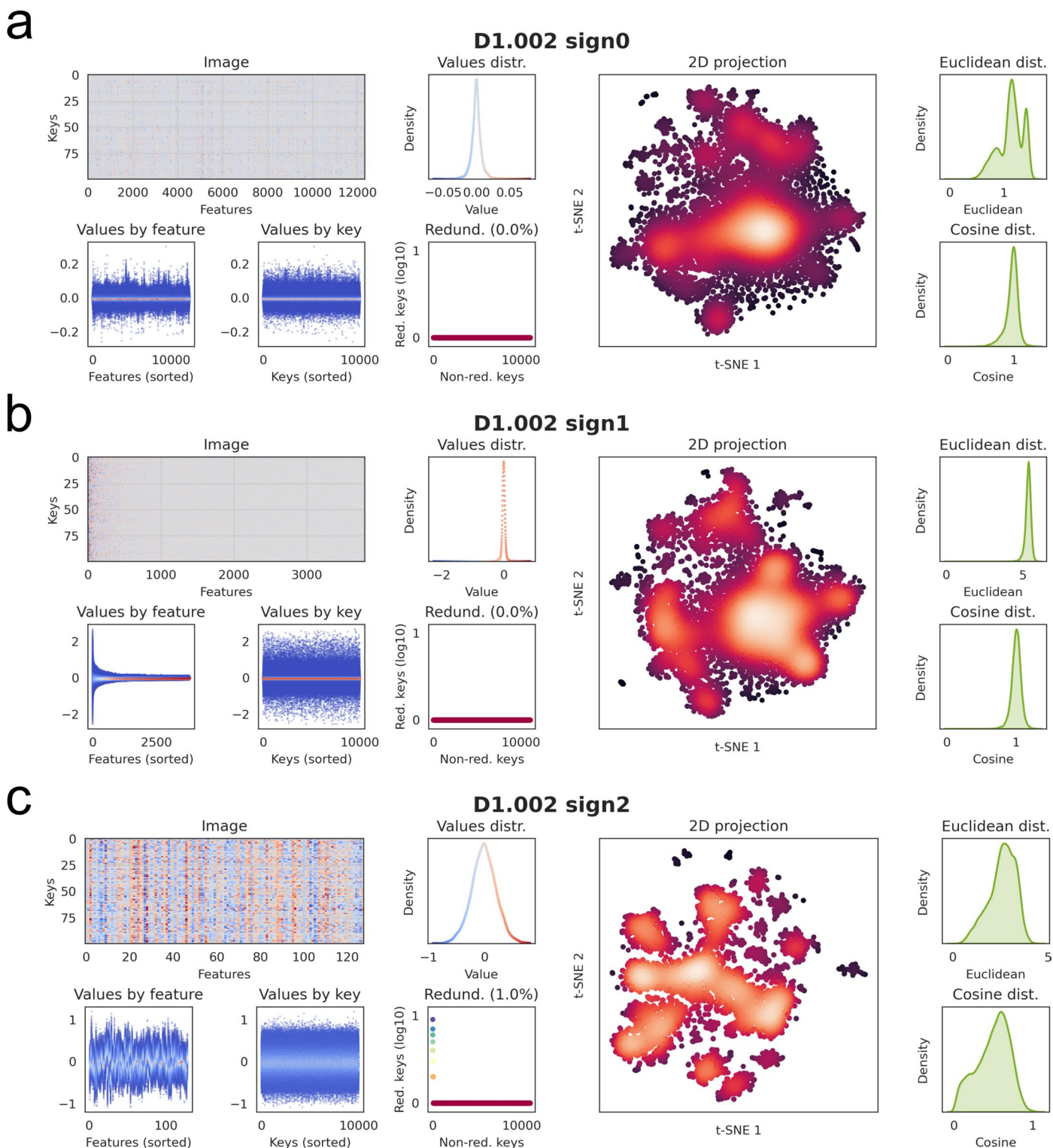
Overview of the Chemical Checker data integration pipeline - Interpreting the results: diagnosis plots.



Extended Data Fig. 2 | Comparison between B1.001 and B1.002. Left: 2D tSNE representation of B1.001 and B1.002 type III signatures having a corresponding type 0 signature in B1.001 and B1.002, respectively. Right: recapitulation of MOA

(B1.001 type 0 signatures, purple) and ATC (E1.001 type 0 signatures, orange) using B1.001 (top) and B1.002 (bottom) type III signatures.

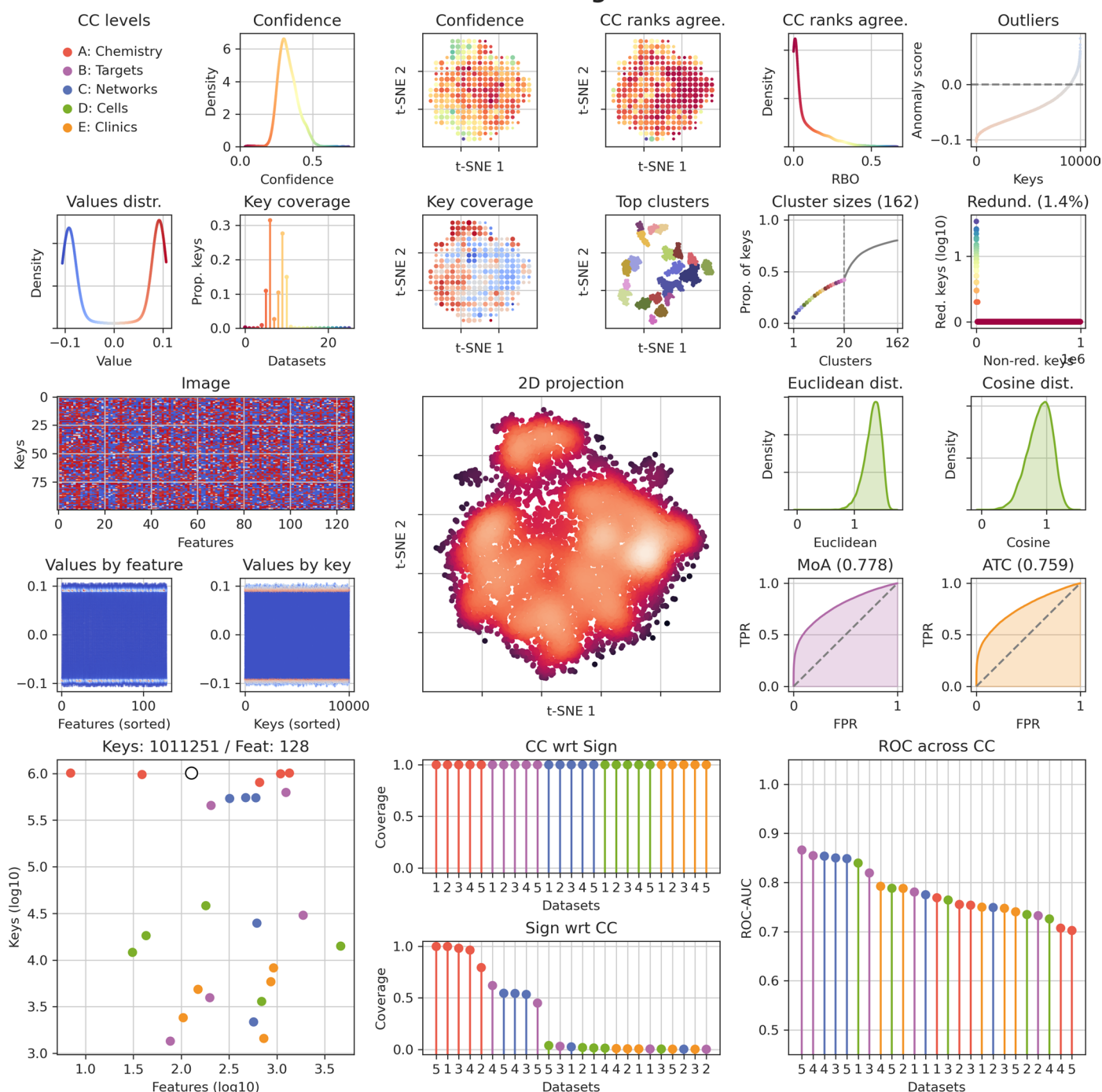
Protocol



Extended Data Fig. 3 | Diagnosis plots for the D1.002 space (please see Anticipated Results: Task 2). **a)** type 0 signatures, **b)** type I signatures and **c)** type II signatures. For further information about diagnosis plots, please see

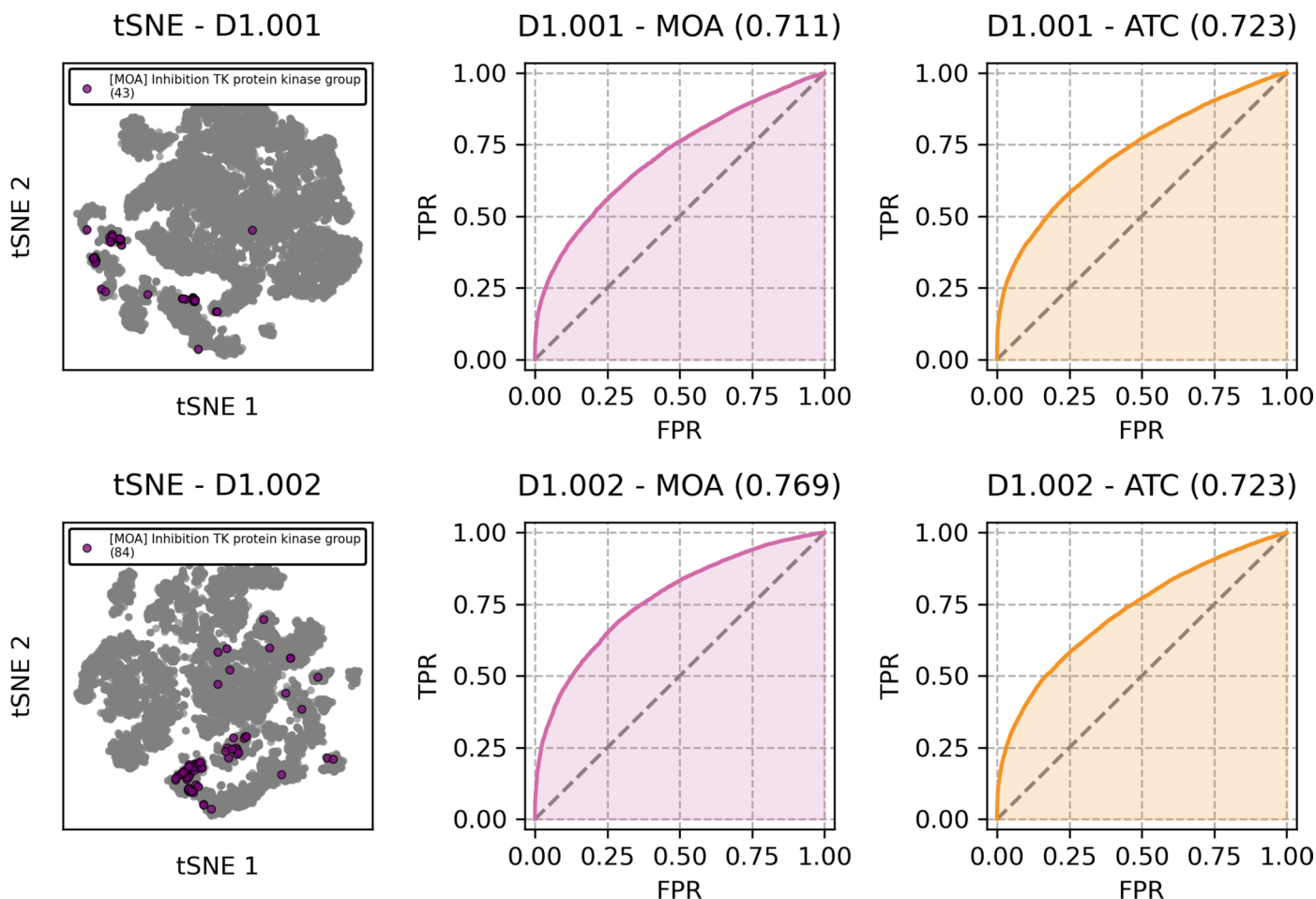
Overview of the Chemical Checker data integration pipeline - Interpreting the results: diagnosis plots.

D1.002 sign3



Extended Data Fig. 4 | Extended diagnosis plots for D1.002 type III signatures (please see **Anticipated Results: Task 2**). For further information about

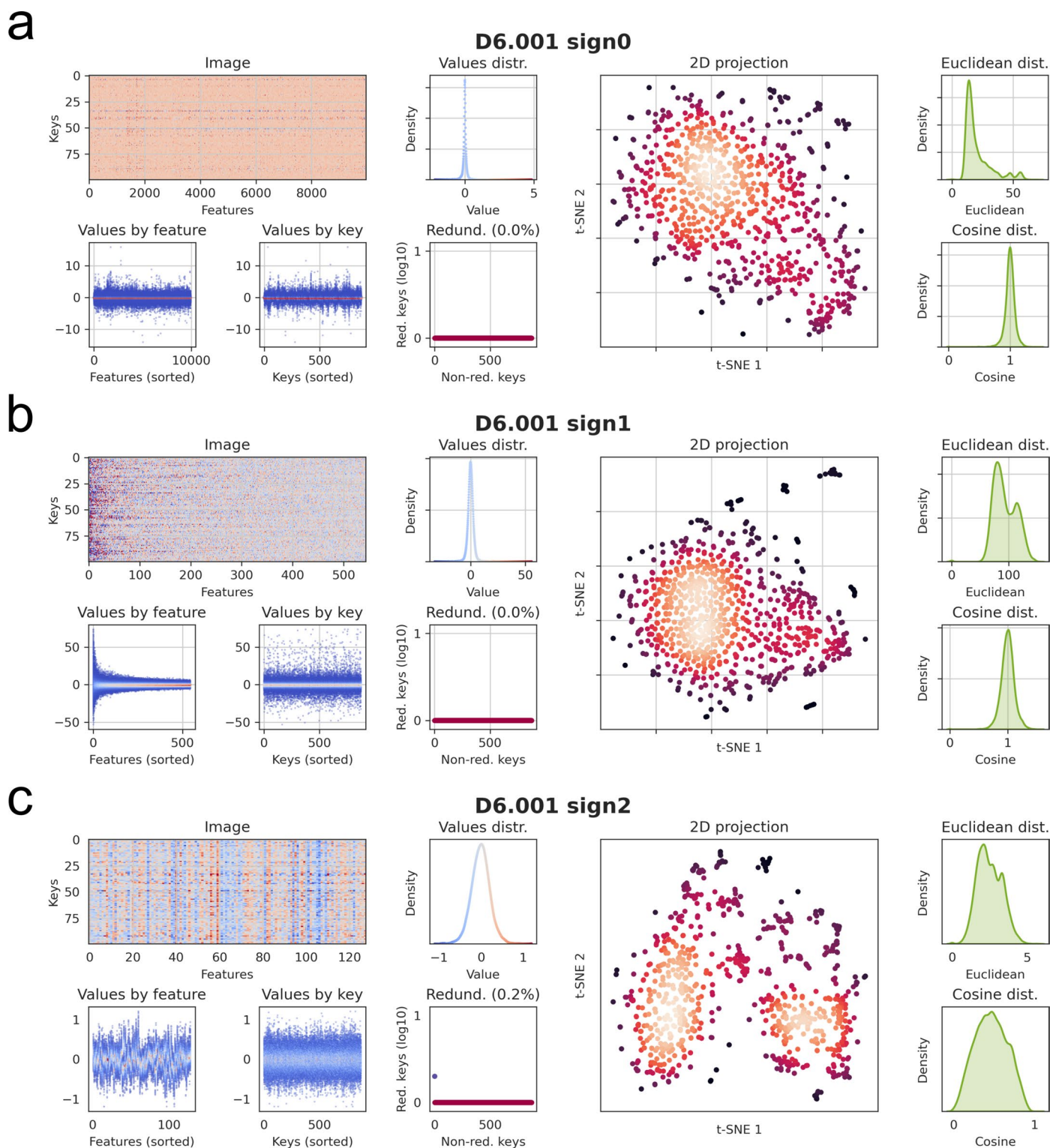
diagnosis plots, please see *Overview of the Chemical Checker data integration pipeline - Interpreting the results: diagnosis plots* and check our Gitlab repository.



Extended Data Fig. 5 | Comparison between D1.001 and D1.002. Left: 2D tSNE representation of D1.001 (top) and D1.002 (bottom) type III signatures having a corresponding type 0 signature in D1.001 and D1.002, respectively. Points highlighted in purple correspond to compounds related to the inhibition of

the TK protein group. Center and right: recapitulation of MOA (B1.001 type 0 signatures, purple) and ATC (E1.001 type 0 signatures, orange) using D1.001 (top) and D1.002 (bottom) type III signatures.

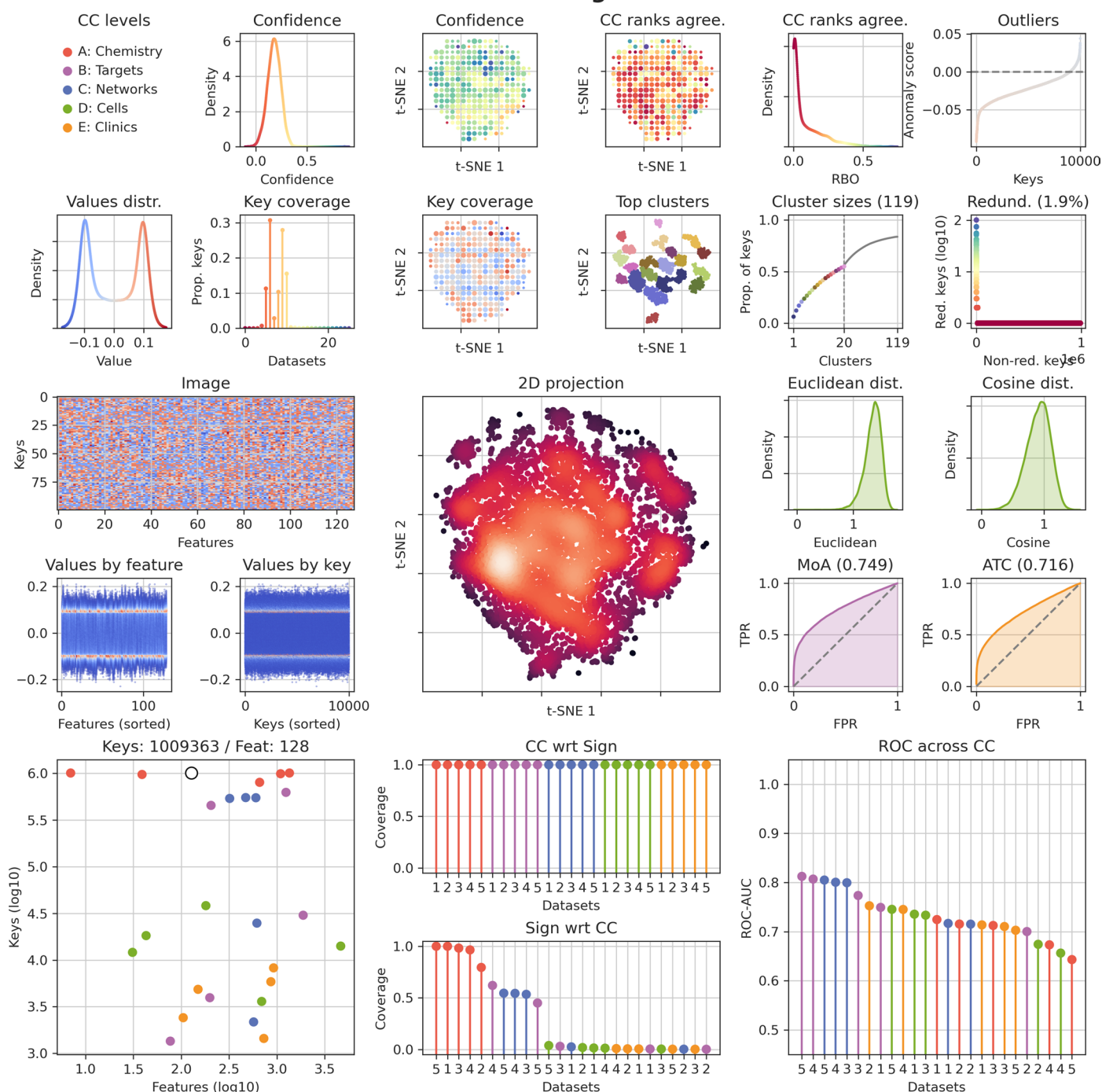
Protocol



Extended Data Fig. 6 | Diagnosis plots for the D6.002 space (please see Anticipated Results: Task 3). a) type 0 signatures, b) type I signatures and c) type II signatures. For further information about diagnosis plots, please see

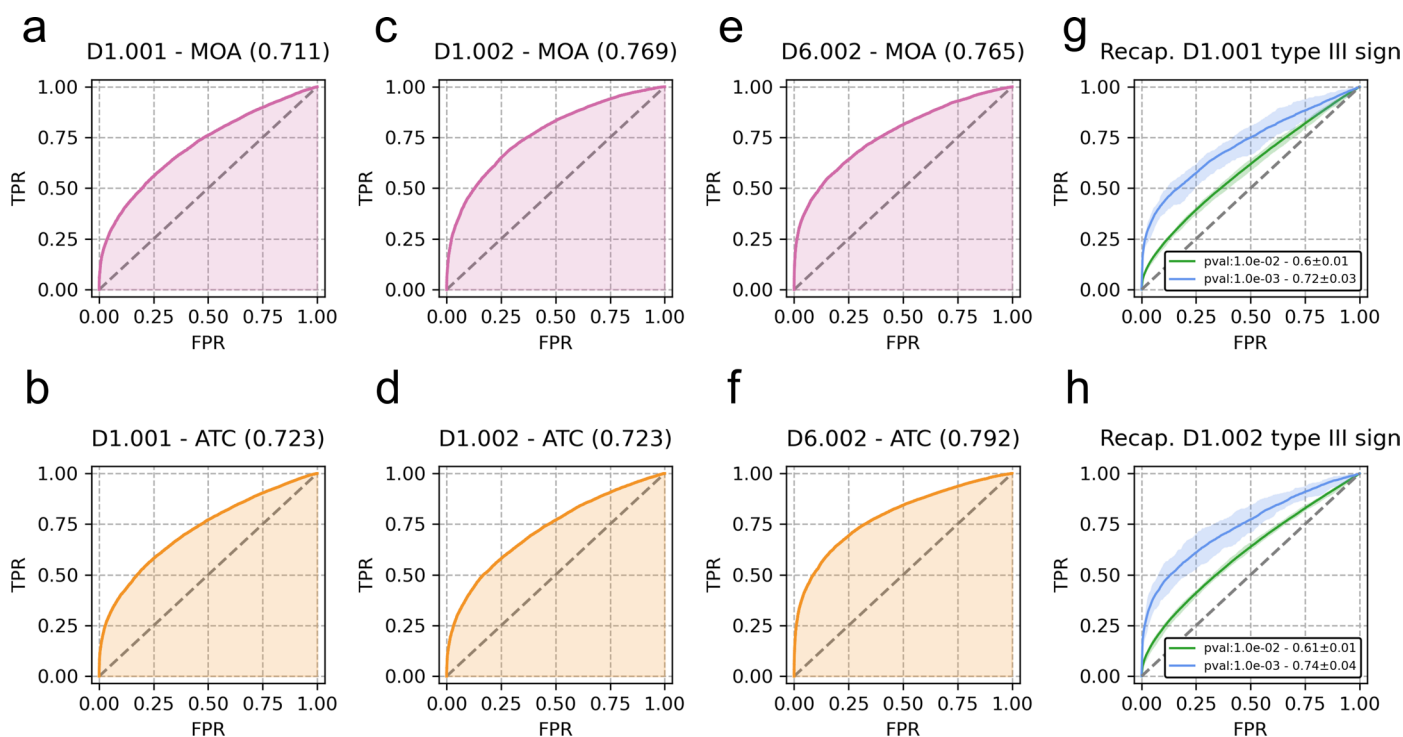
Overview of the Chemical Checker data integration pipeline - Interpreting the results: diagnosis plots.

D6.001 sign3



Extended Data Fig. 7 | Extended diagnosis plots for D6.002 type III signatures (please see *Anticipated Results: Task 3*). For further information about

diagnosis plots, please see *Overview of the Chemical Checker data integration pipeline · Interpreting the results: diagnosis plots* and check our Gitlab repository.



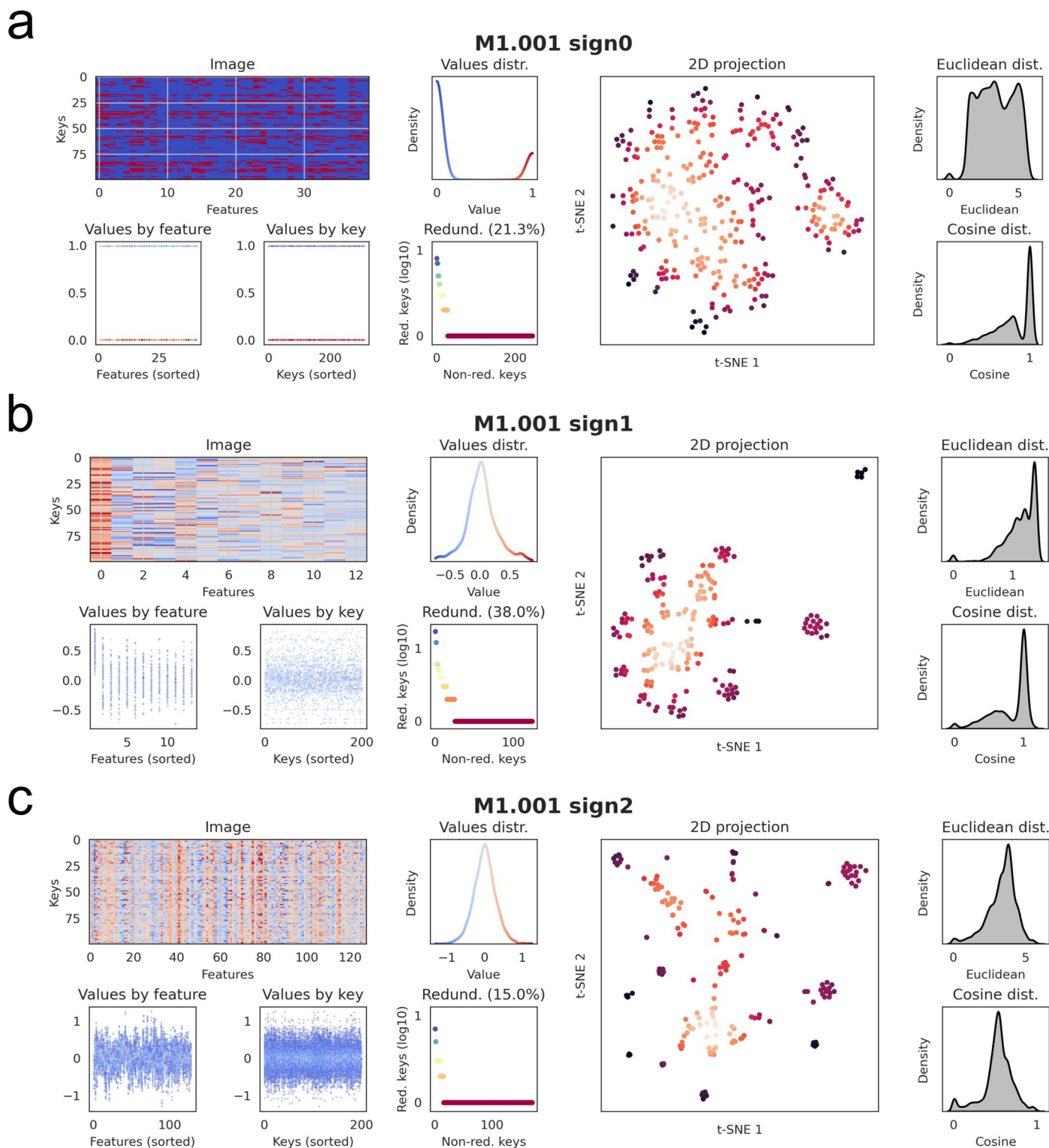
Extended Data Fig. 8 | Comparison between D1.001, D1.002 and D6.001.

a Recapitulation of MoA (B1.001 type 0 signatures, purple) using D1.001 type III signatures. **b** Recapitulation of ATC (E1.001 type 0 signatures, orange) using D1.001 type III signatures. **c** Recapitulation of MoA (B1.001 type 0 signatures, purple) using D1.002 type III signatures. **d** Recapitulation of ATC (E1.001 type 0 signatures, orange) using D1.002 type III signatures. **e** Recapitulation

of MoA (B1.001 type 0 signatures, purple) using D6.001 type III signatures.

f Recapitulation of ATC (E1.001 type 0 signatures, orange) using D6.001 type III signatures. **g** Recapitulation of kNN at D1.001 type III signature level using D6.001 type III signatures at p-values 0.01 and 0.001. **h** Recapitulation of NN at D1.002 type III signature level using D6.001 type III signatures at p-values 0.01 and 0.001.

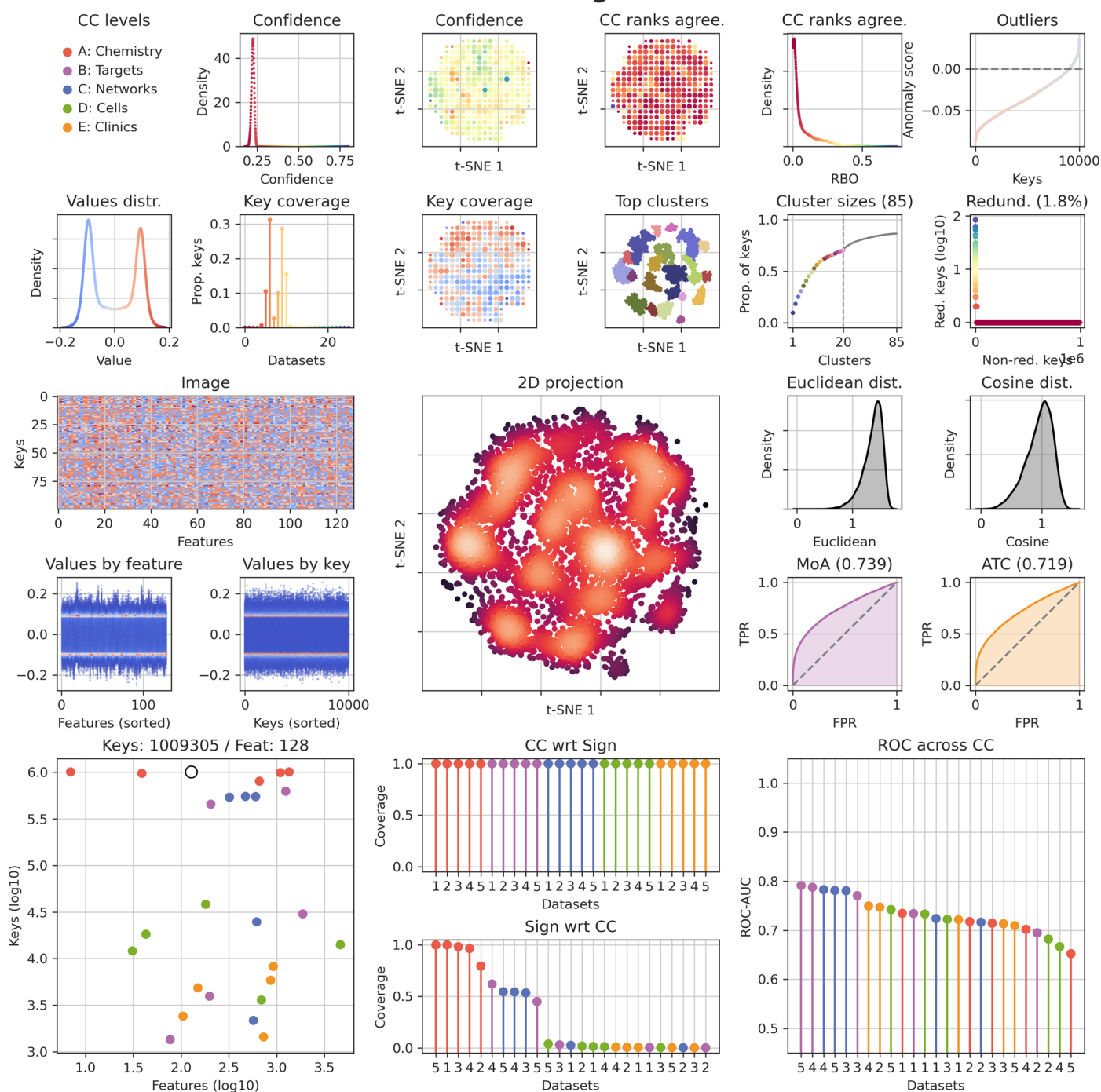
Protocol



Extended Data Fig. 9 | Diagnosis plots for the M1.001 space (please see Anticipated Results: Task 4). **a)** type 0 signatures, **b)** type I signatures and **c)** type II signatures. For further information about diagnosis plots, please see

Overview of the Chemical Checker data integration pipeline - Interpreting the results: diagnosis plots.

M1.001 sign3



Extended Data Fig. 10 | Extended diagnosis plots for M1.001 type III signatures (please see *Anticipated Results: Task 4*). For further information about diagnosis plots, please see *Overview of the Chemical Checker data*

integration pipeline · Interpreting the results: diagnosis plots and check our Gitlab repository.

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Sample size	Data sample sizes were determined on the basis of publicly available bioactivity data.
Data exclusions	No data were excluded.
Replication	The CC Protocol has an inherent stochastic behaviour. For this reason, in absolute terms, numerical signatures may differ among distinct runs of the pipeline. However, relative relationships and similarities (i.e. if two compounds are similar or dissimilar in terms of their signatures) among different runs are retained.
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Blinding	Researchers were not blinded during the processing of experimental samples.

Reporting for specific materials, systems and methods

We require information from authors about some types of materials, experimental systems and methods used in many studies. Here, indicate whether each material, system or method listed is relevant to your study. If you are not sure if a list item applies to your research, read the appropriate section before selecting a response.

Materials & experimental systems

n/a	Involved in the study
☒	☐ Antibodies
☒	☐ Eukaryotic cell lines
☒	☐ Palaeontology and archaeology
☒	☐ Animals and other organisms
☒	☐ Clinical data
☒	☐ Dual use research of concern
☒	☐ Plants

Methods

n/a	Involved in the study
☒	☐ ChIP-seq
☒	☐ Flow cytometry
☒	☐ MRI-based neuroimaging

Plants

Seed stocks

Report on the source of all seed stocks or other plant material used. If applicable, state the seed stock centre and catalogue number. If plant specimens were collected from the field, describe the collection location, date and sampling procedures.

Novel plant genotypes

Describe the methods by which all novel plant genotypes were produced. This includes those generated by transgenic approaches, gene editing, chemical/radiation-based mutagenesis and hybridization. For transgenic lines, describe the transformation method, the number of independent lines analyzed and the generation upon which experiments were performed. For gene-edited lines, describe the editor used, the endogenous sequence targeted for editing, the targeting guide RNA sequence (if applicable) and how the editor was applied.

Authentication

Describe any authentication procedures for each seed stock used or novel genotype generated. Describe any experiments used to assess the effect of a mutation and, where applicable, how potential secondary effects (e.g. second site T-DNA insertions, mosaicism, off-target gene editing) were examined.